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(54) **POINT MANAGEMENT SYSTEM, POINT MANAGEMENT METHOD, AND NON-TRANSITORY RECORDING MEDIUM**

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(57) **ABSTRACT**

A point management system includes circuitry to issue first identification information identifying a point for a consumer to obtain a coupon, register the first identification information in first association information, and calculate an actual value of a carbon dioxide emission from an amount of activity for a printed material. The printed material bears a symbol including the first identification information. The circuitry calculates the number of points based on the actual value, and registers the calculated points in the first association information in association with the first identification information.

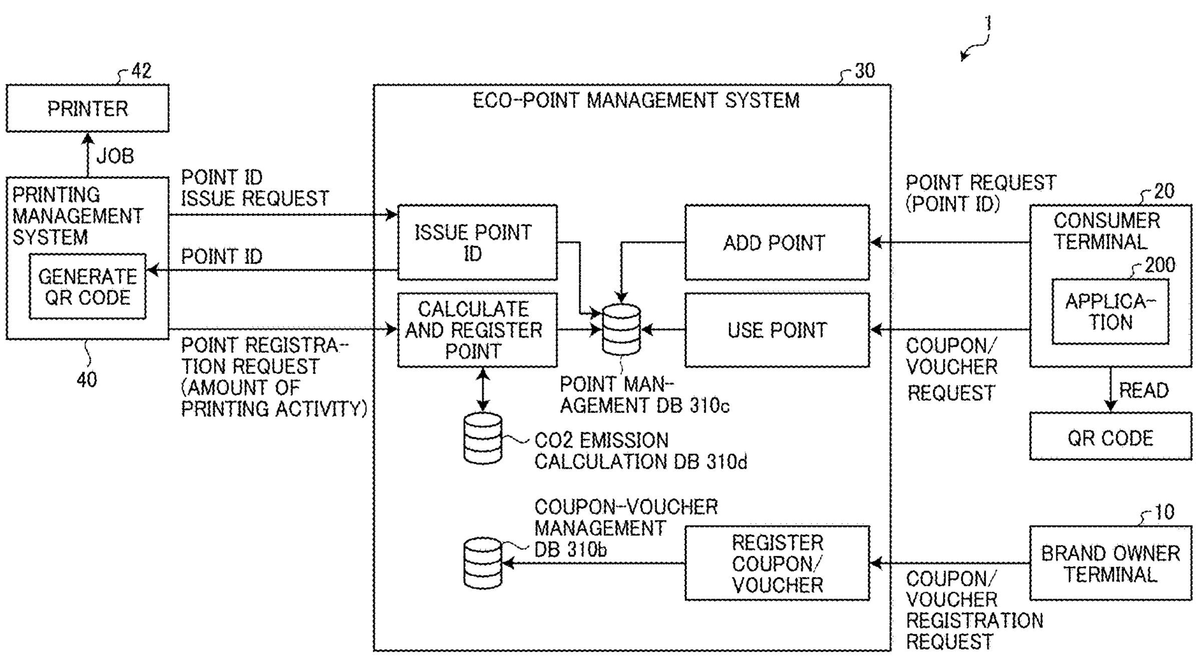


FIG. 1

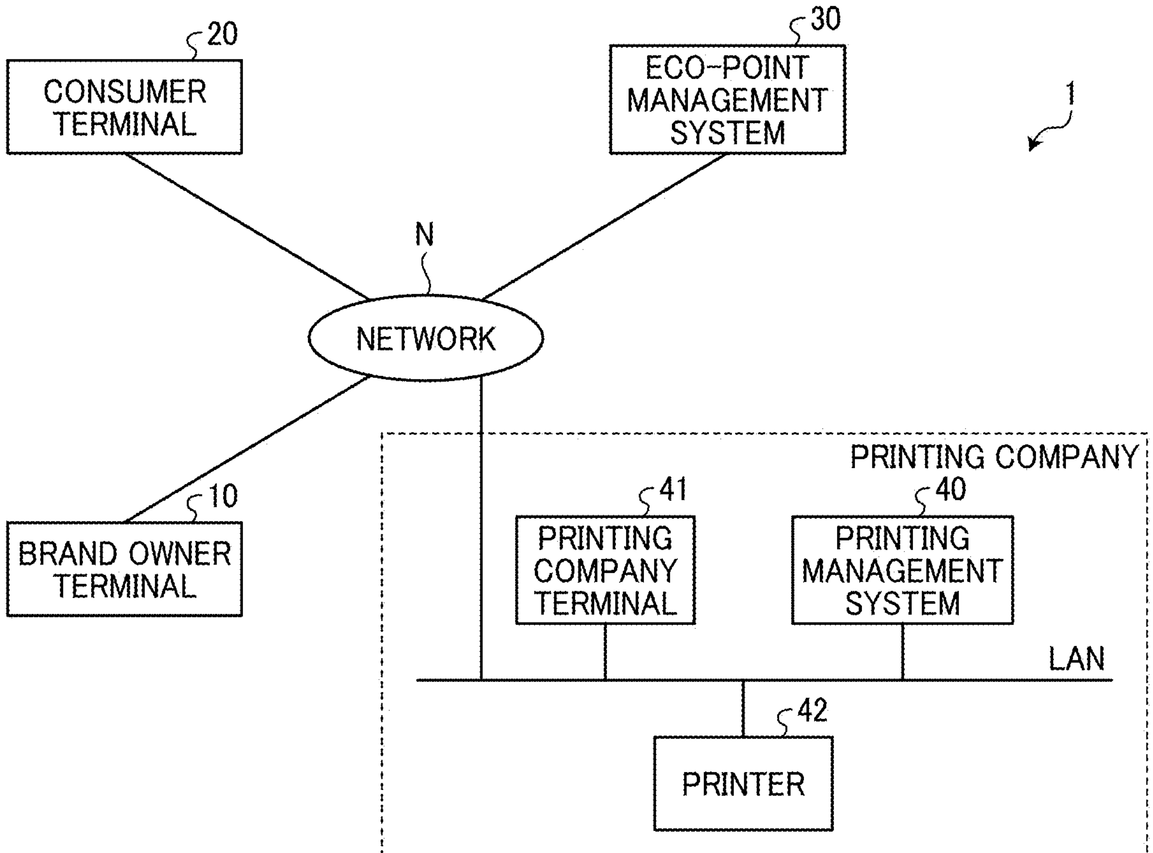

1
20
CONSUMER TERMINAL
30
ECO-POINT MANAGEMENT SYSTEM
N
NETWORK
10
BRAND OWNER TERMINAL
PRINTING COMPANY
41
PRINTING COMPANY TERMINAL
40
PRINTING MANAGEMENT SYSTEM
LAN
42
PRINTER

FIG. 2

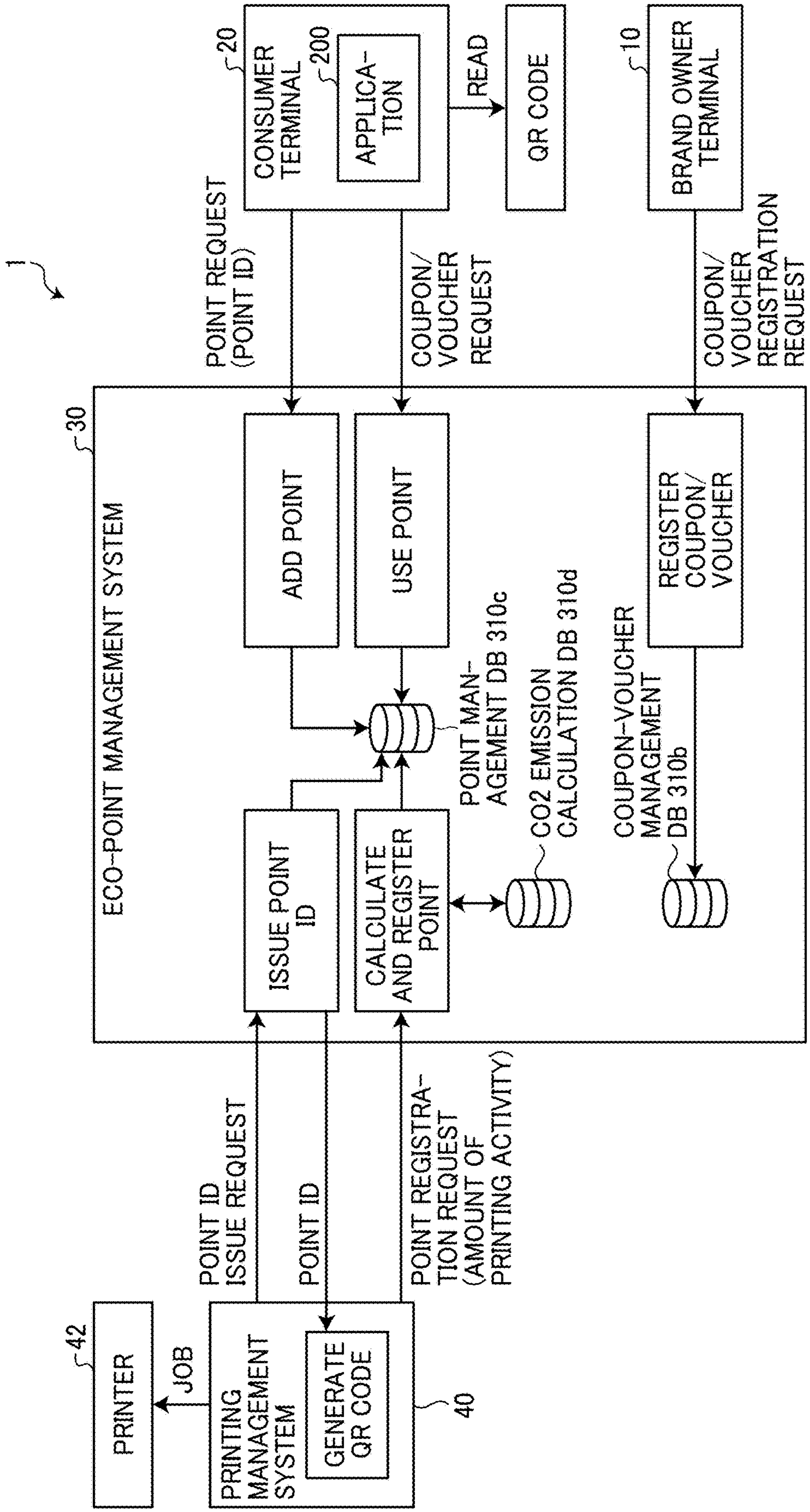

1
20
CONSUMER TERMINAL
200
APPLICATION
READ
QR CODE
10
BRAND OWNER TERMINAL
POINT REQUEST (POINT ID)
COUPON/ VOUCHER REQUEST
COUPON/ VOUCHER REGISTRATION REQUEST
30
ECO-POINT MANAGEMENT SYSTEM
ADD POINT
USE POINT
REGISTER COUPON/ VOUCHER
POINT MANAGEMENT DB 310c
CO2 EMISSION CALCULATION DB 310d
COUPON-VOUCHER MANAGEMENT DB 310b
ISSUE POINT ID
CALCULATE AND REGISTER POINT
POINT ID ISSUE REQUEST
POINT ID
POINT REGISTRATION REQUEST (AMOUNT OF PRINTING ACTIVITY)
42
PRINTER
JOB
PRINTING MANAGEMENT SYSTEM
GENERATE QR CODE
40

FIG. 3

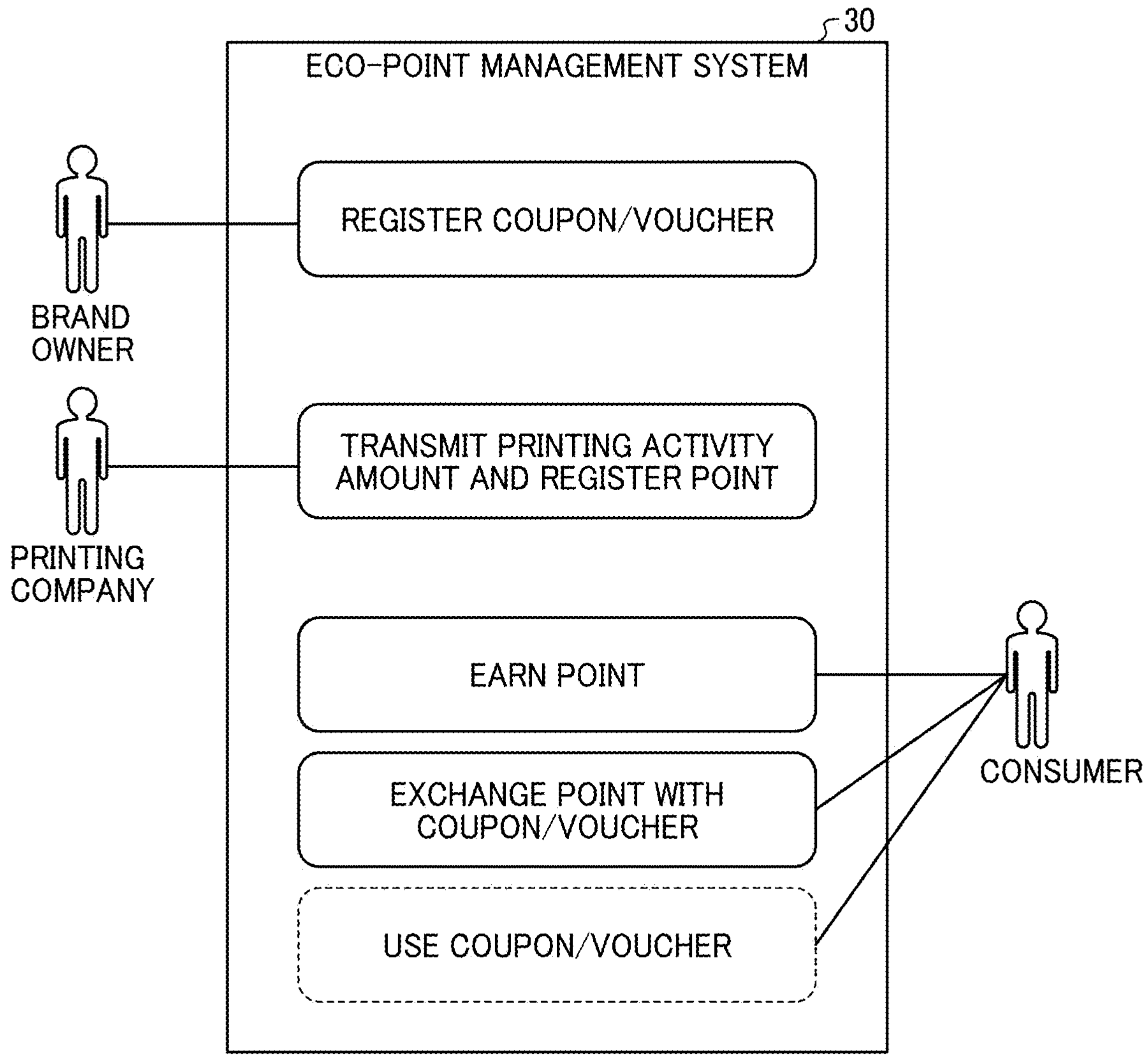

30
ECO-POINT MANAGEMENT SYSTEM
REGISTER COUPON/VOUCHER
BRAND OWNER
TRANSMIT PRINTING ACTIVITY AMOUNT AND REGISTER POINT
PRINTING COMPANY
EARN POINT
EXCHANGE POINT WITH COUPON/VOUCHER
USE COUPON/VOUCHER
CONSUMER

FIG. 4

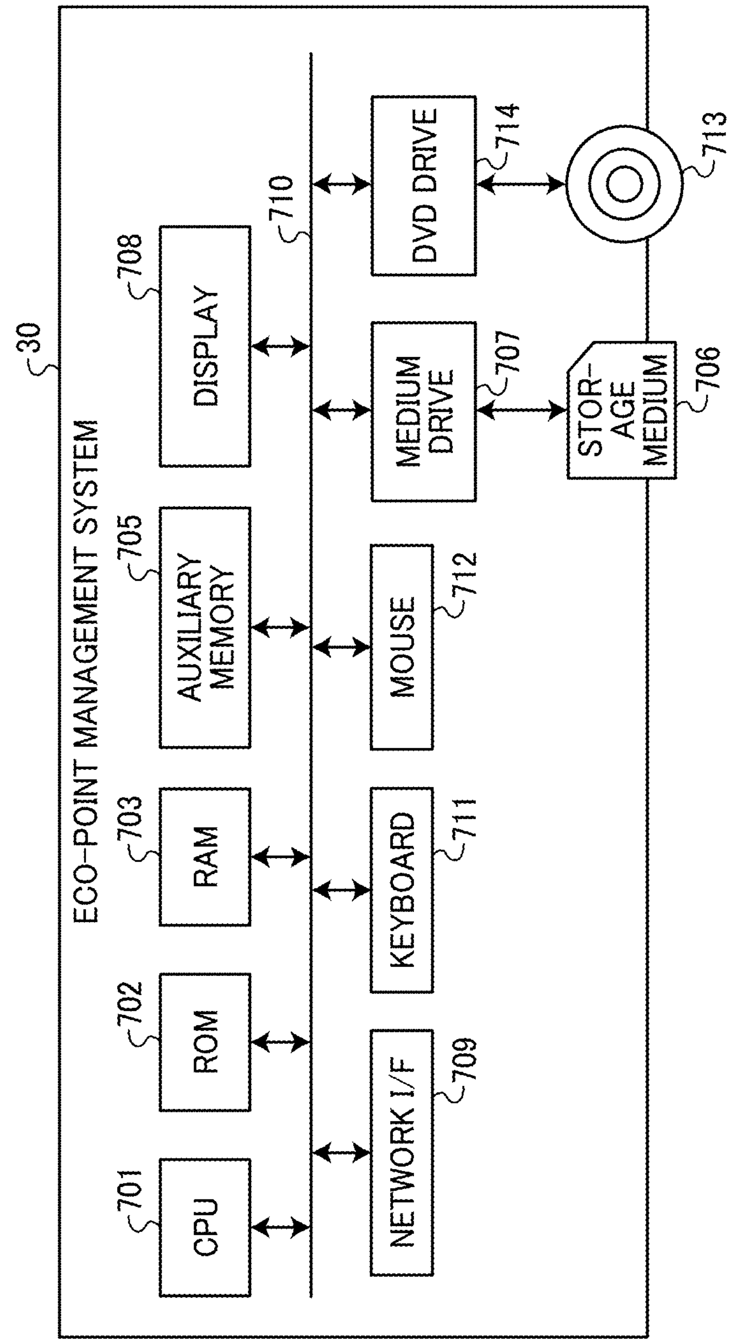

30
ECO-POINT MANAGEMENT SYSTEM
CPU
701
ROM
702
RAM
703
AUXILIARY MEMORY
705
DISPLAY
708
710
NETWORK I/F
709
KEYBOARD
711
MOUSE
712
MEDIUM DRIVE
707
DVD DRIVE
714
STOR-AGE MEDIUM
706
713

FIG. 5

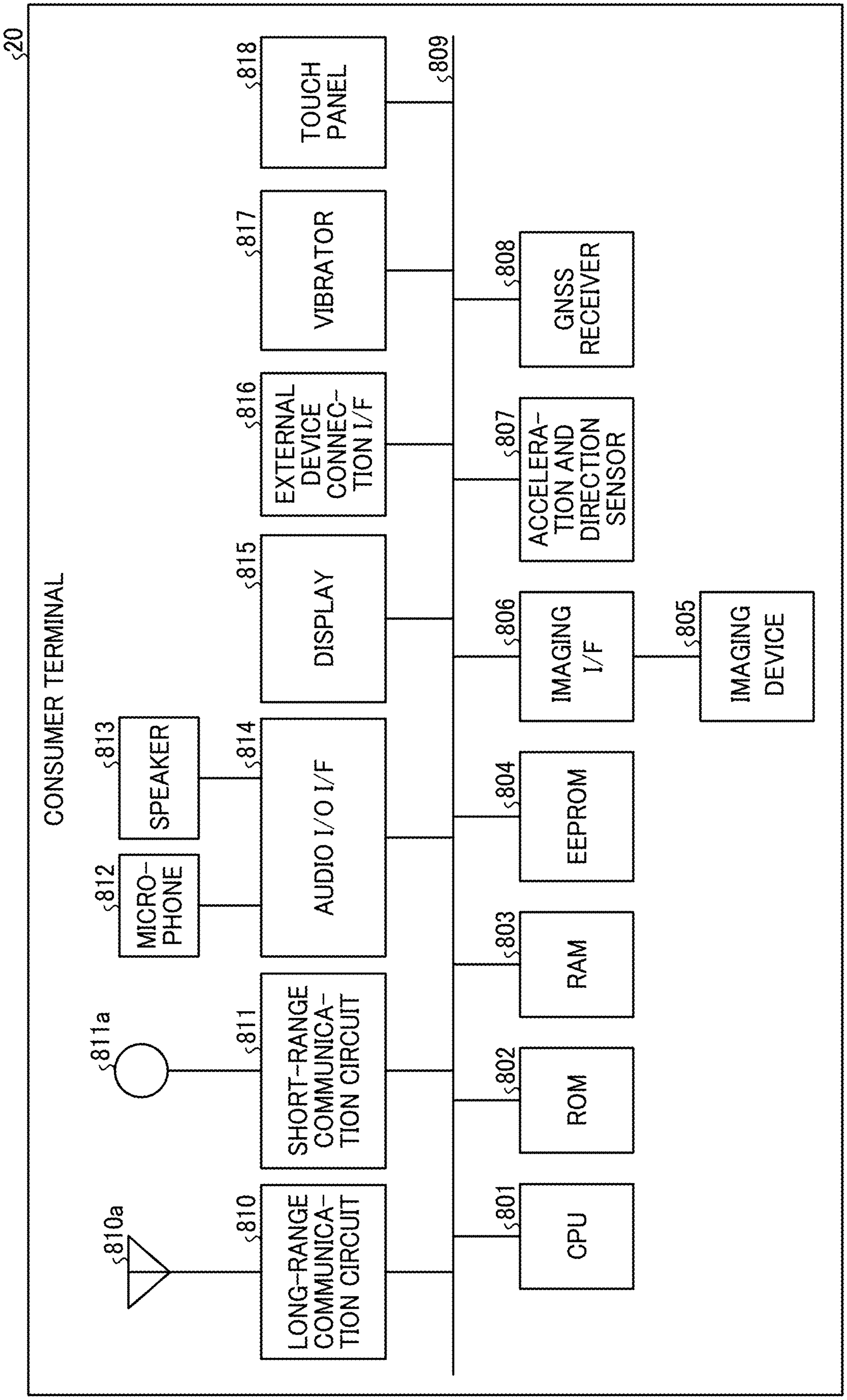

20
CONSUMER TERMINAL
810a
810
LONG-RANGE COMMUNICA-TION CIRCUIT
811a
811
SHORT-RANGE COMMUNICA-TION CIRCUIT
812
MICRO-PHONE
813
SPEAKER
814
AUDIO I/O I/F
815
DISPLAY
816
EXTERNAL DEVICE CONNEC-TION I/F
817
VIBRATOR
818
TOUCH PANEL
809
801
CPU
802
ROM
803
RAM
804
EEPROM
806
IMAGING I/F
805
IMAGING DEVICE
807
ACCELERA-TION AND DIRECTION SENSOR
808
GNSS RECEIVER

FIG. 6

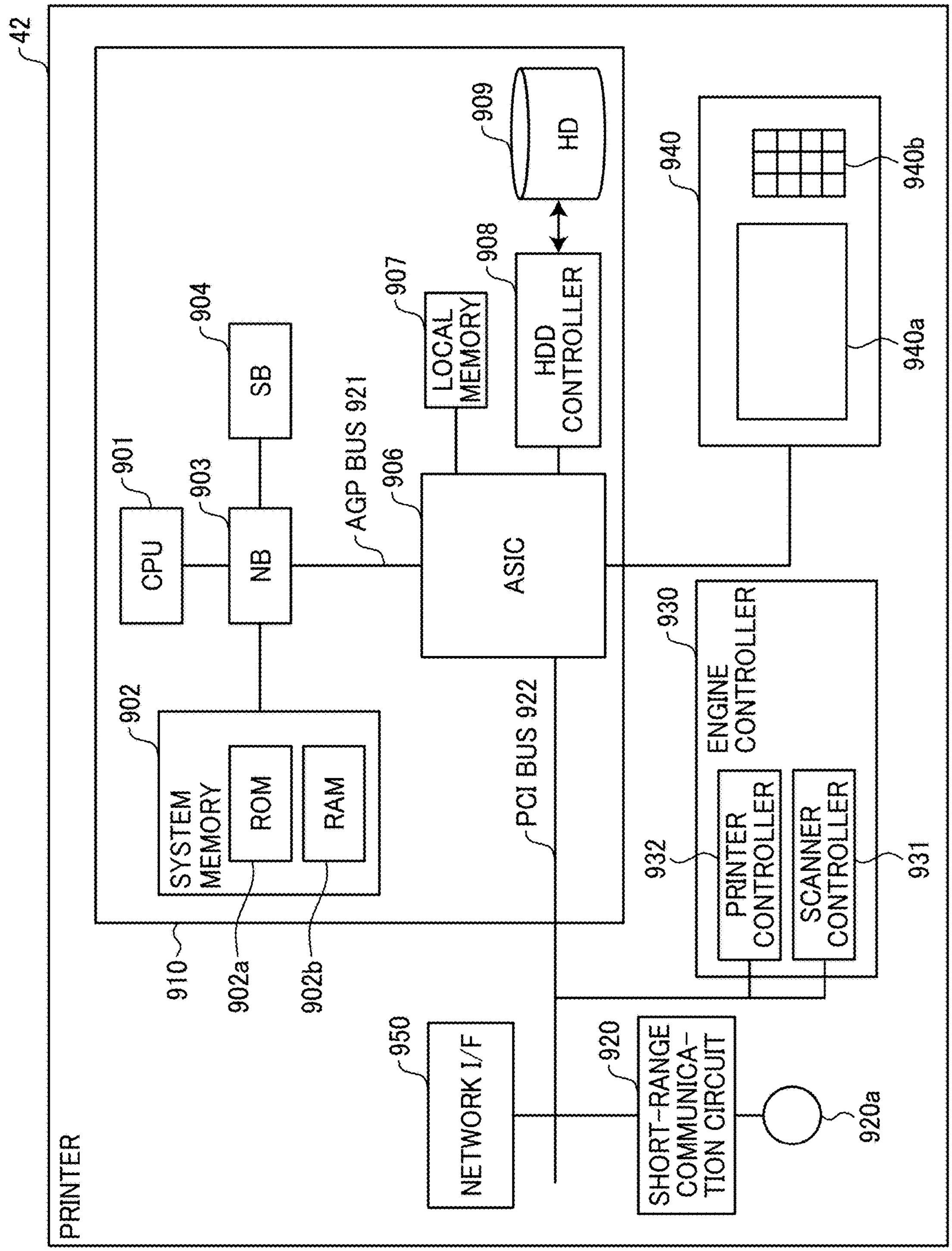

42
PRINTER
910
SYSTEM MEMORY
902
ROM
902a
RAM
902b
CPU
901
NB
903
SB
904
AGP BUS 921
ASIC
906
LOCAL MEMORY
907
HDD CONTROLLER
908
HD
909
PCI BUS 922
940
940a
940b
ENGINE CONTROLLER
930
PRINTER CONTROLLER
932
SCANNER CONTROLLER
931
NETWORK I/F
950
SHORT-RANGE COMMUNICA-TION CIRCUIT
920
920a

FIG. 7

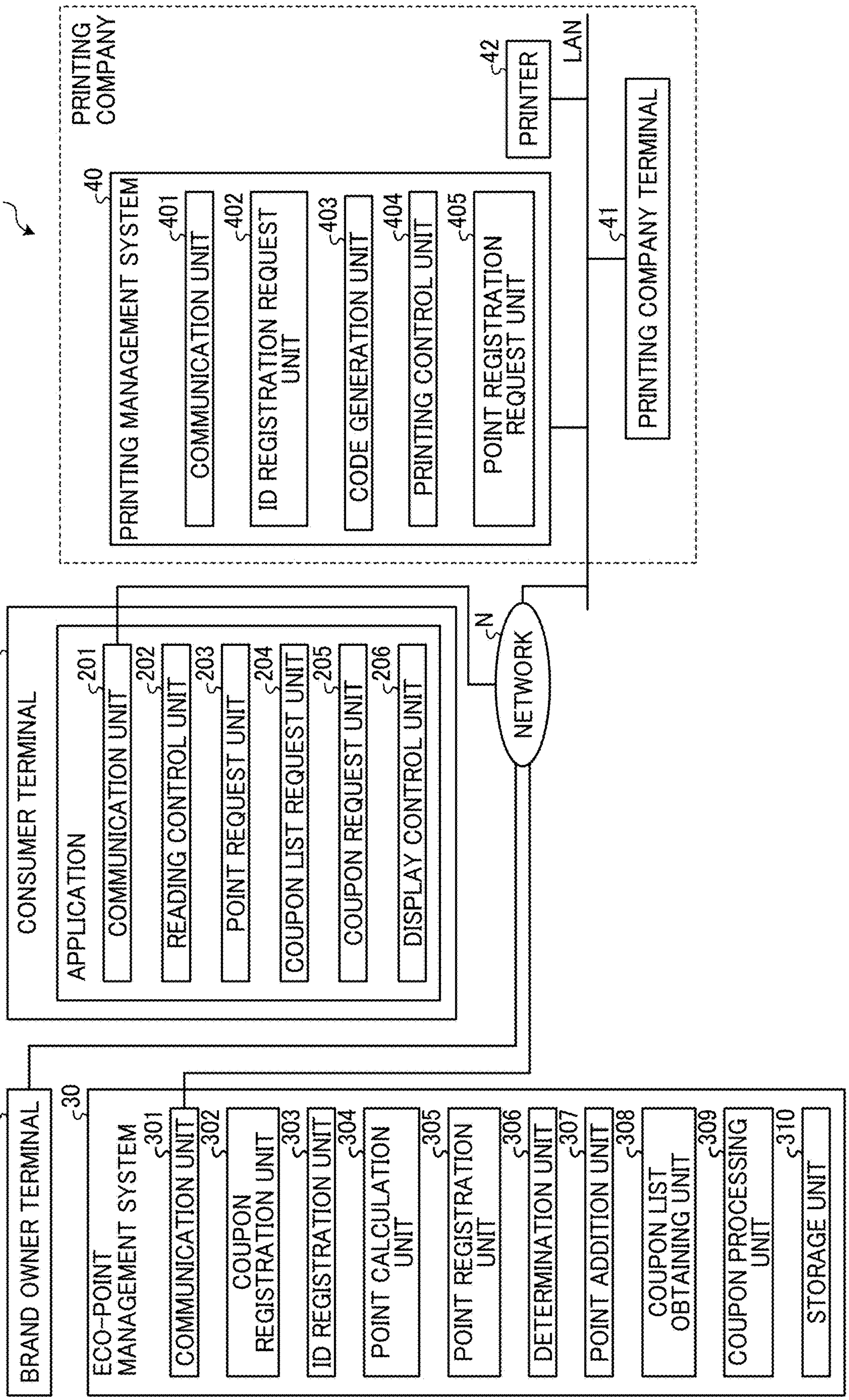

1
BRAND OWNER TERMINAL
10
ECO-POINT MANAGEMENT SYSTEM
30
COMMUNICATION UNIT
301
COUPON REGISTRATION UNIT
302
ID REGISTRATION UNIT
303
POINT CALCULATION UNIT
304
POINT REGISTRATION UNIT
305
DETERMINATION UNIT
306
POINT ADDITION UNIT
307
COUPON LIST OBTAINING UNIT
308
COUPON PROCESSING UNIT
309
STORAGE UNIT
310
CONSUMER TERMINAL
20
APPLICATION
COMMUNICATION UNIT
201
READING CONTROL UNIT
202
POINT REQUEST UNIT
203
COUPON LIST REQUEST UNIT
204
COUPON REQUEST UNIT
205
DISPLAY CONTROL UNIT
206
NETWORK
N
PRINTING COMPANY
PRINTING MANAGEMENT SYSTEM
40
COMMUNICATION UNIT
401
ID REGISTRATION REQUEST UNIT
402
CODE GENERATION UNIT
403
PRINTING CONTROL UNIT
404
POINT REGISTRATION REQUEST UNIT
405
PRINTER
42
LAN
PRINTING COMPANY TERMINAL
41

FIG. 8

310a

USER MANAGEMENT DB
USER ID
PASSWORD
POINT BALANCE

310c

POINT MANAGEMENT DB
POINT ID
BRAND OWNER
NUMBER OF POINTS
USER ID

310b

COUPON-VOUCHER MANAGEMENT DB
BRAND OWNER
PRODUCT NAME
REQUIRED POINT
STORES ACCEPTING COUPON/VOUCHER
NUMBER OF COUPON/VOUCHER
EXPIRATION DATE
CONDITION

310d

CO_2 EMISSION CALCULATION DB
EMISSION INTENSITY

FIG. 9

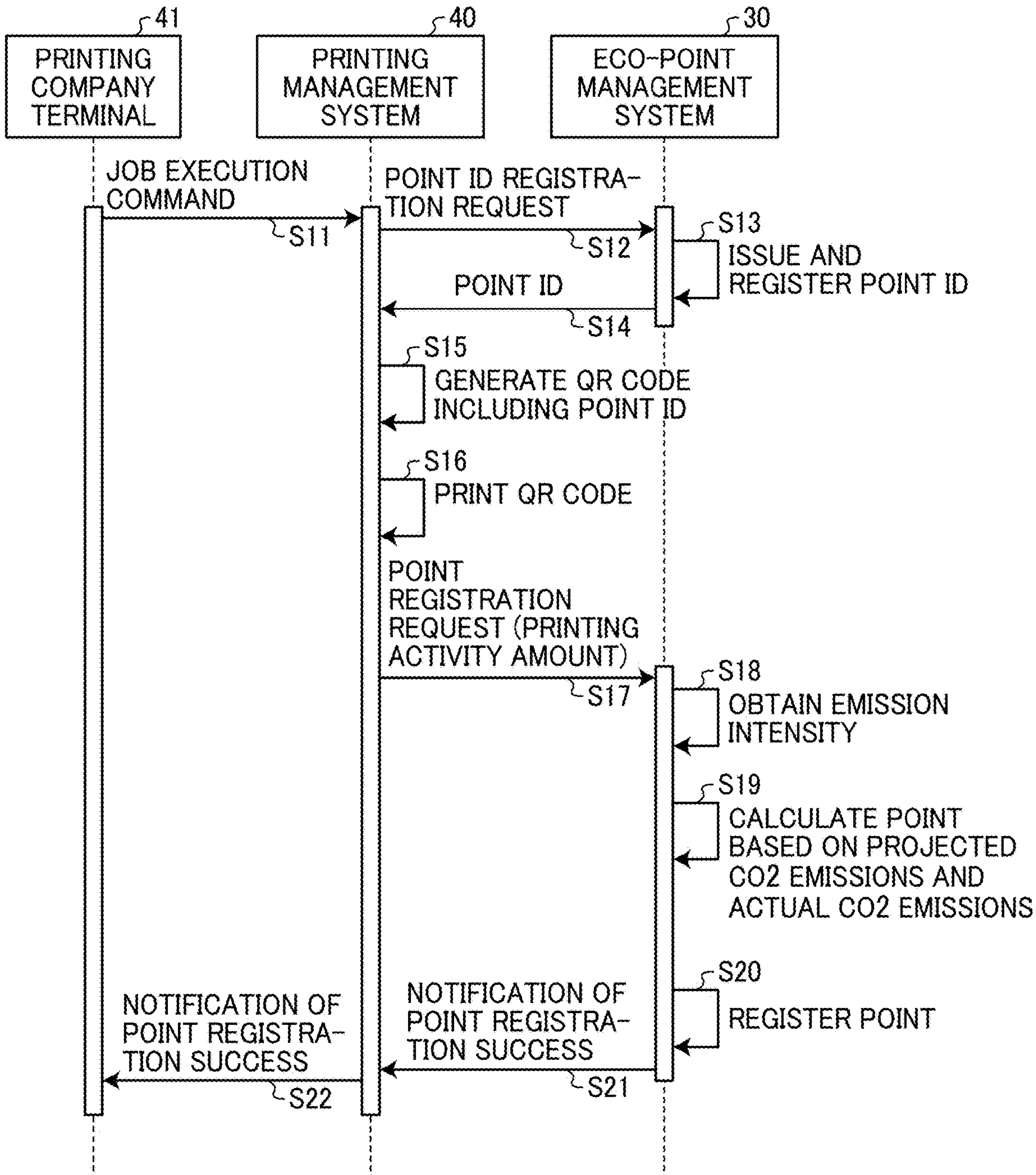

41
PRINTING COMPANY TERMINAL
40
PRINTING MANAGEMENT SYSTEM
30
ECO-POINT MANAGEMENT SYSTEM
JOB EXECUTION COMMAND
S11
POINT ID REGISTRATION REQUEST
S12
S13
ISSUE AND REGISTER POINT ID
POINT ID
S14
S15
GENERATE QR CODE INCLUDING POINT ID
S16
PRINT QR CODE
POINT REGISTRATION REQUEST (PRINTING ACTIVITY AMOUNT)
S17
S18
OBTAIN EMISSION INTENSITY
S19
CALCULATE POINT BASED ON PROJECTED CO2 EMISSIONS AND ACTUAL CO2 EMISSIONS
S20
REGISTER POINT
NOTIFICATION OF POINT REGISTRATION SUCCESS
S21
NOTIFICATION OF POINT REGISTRATION SUCCESS
S22

FIG. 10

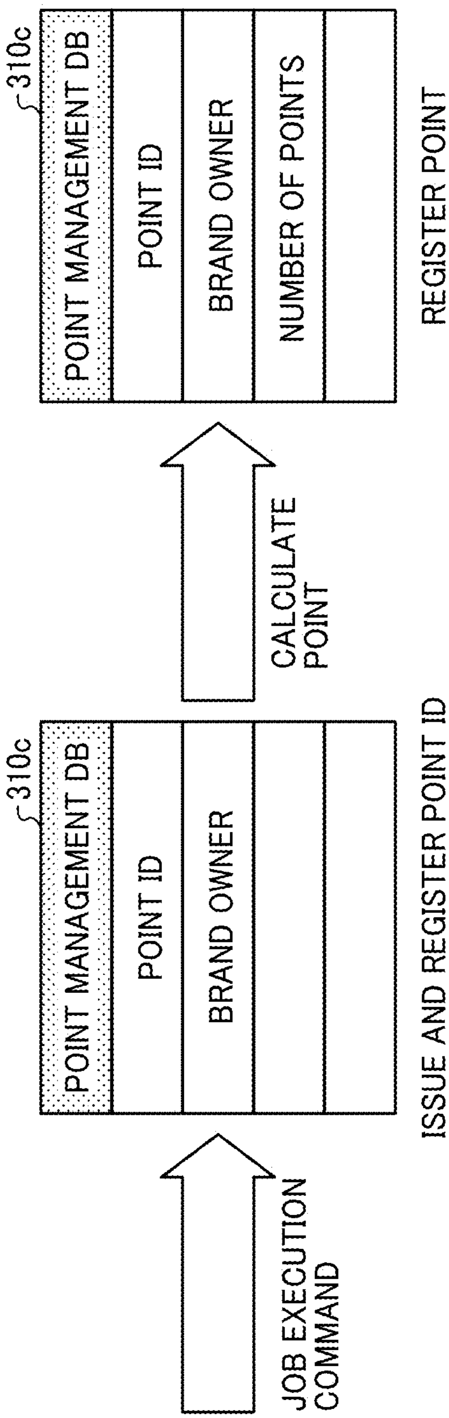

JOB EXECUTION COMMAND
310c
POINT MANAGEMENT DB
POINT ID
BRAND OWNER
ISSUE AND REGISTER POINT ID
CALCULATE POINT
310c
POINT MANAGEMENT DB
POINT ID
BRAND OWNER
NUMBER OF POINTS
REGISTER POINT

FIG. 11

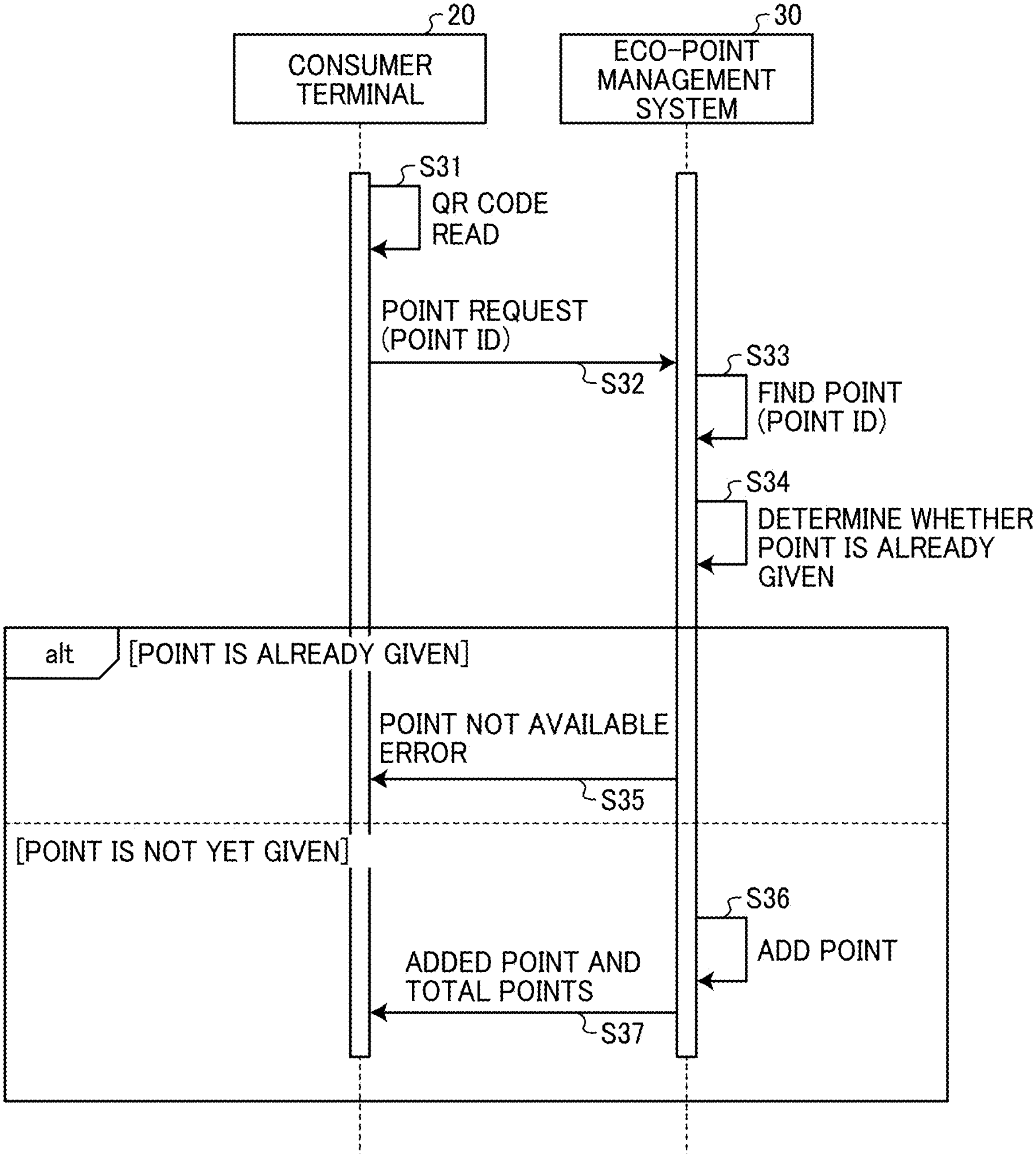

20
CONSUMER TERMINAL
30
ECO-POINT MANAGEMENT SYSTEM
S31
QR CODE READ
POINT REQUEST (POINT ID)
S32
S33
FIND POINT (POINT ID)
S34
DETERMINE WHETHER POINT IS ALREADY GIVEN
alt
[POINT IS ALREADY GIVEN]
POINT NOT AVAILABLE ERROR
S35
[POINT IS NOT YET GIVEN]
S36
ADD POINT
ADDED POINT AND TOTAL POINTS
S37

FIG. 12

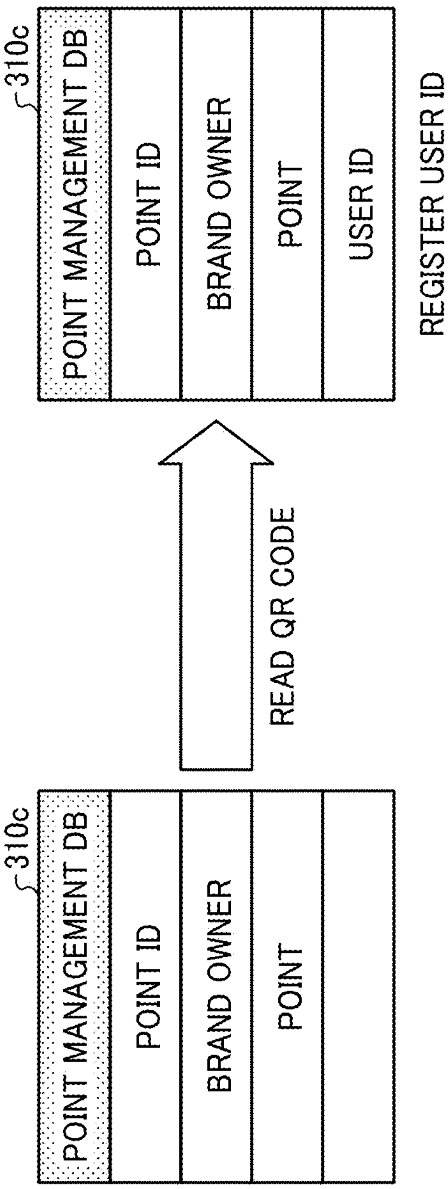

310c
POINT MANAGEMENT DB
POINT ID
BRAND OWNER
POINT
READ QR CODE
310c
POINT MANAGEMENT DB
POINT ID
BRAND OWNER
POINT
USER ID
REGISTER USER ID

FIG. 13

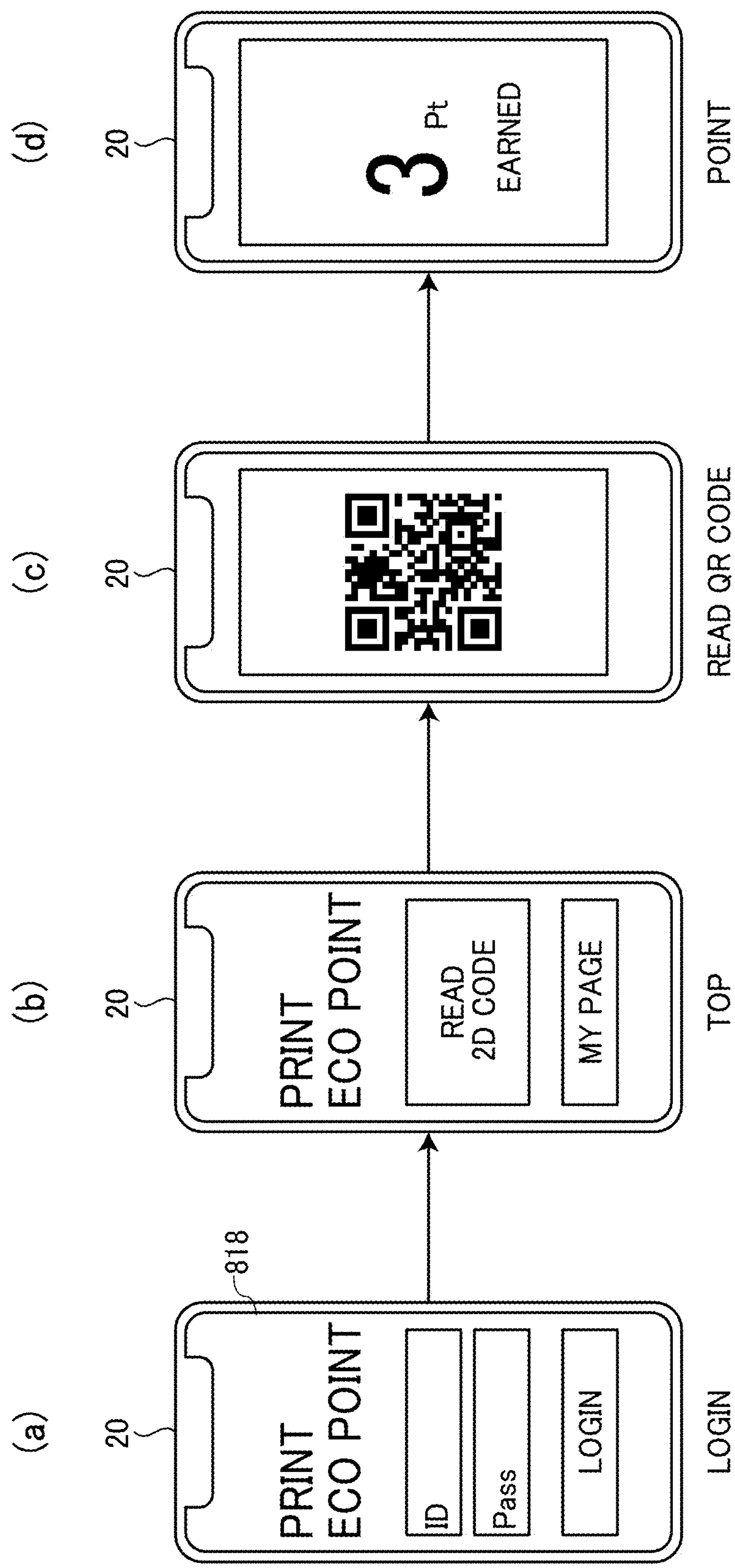
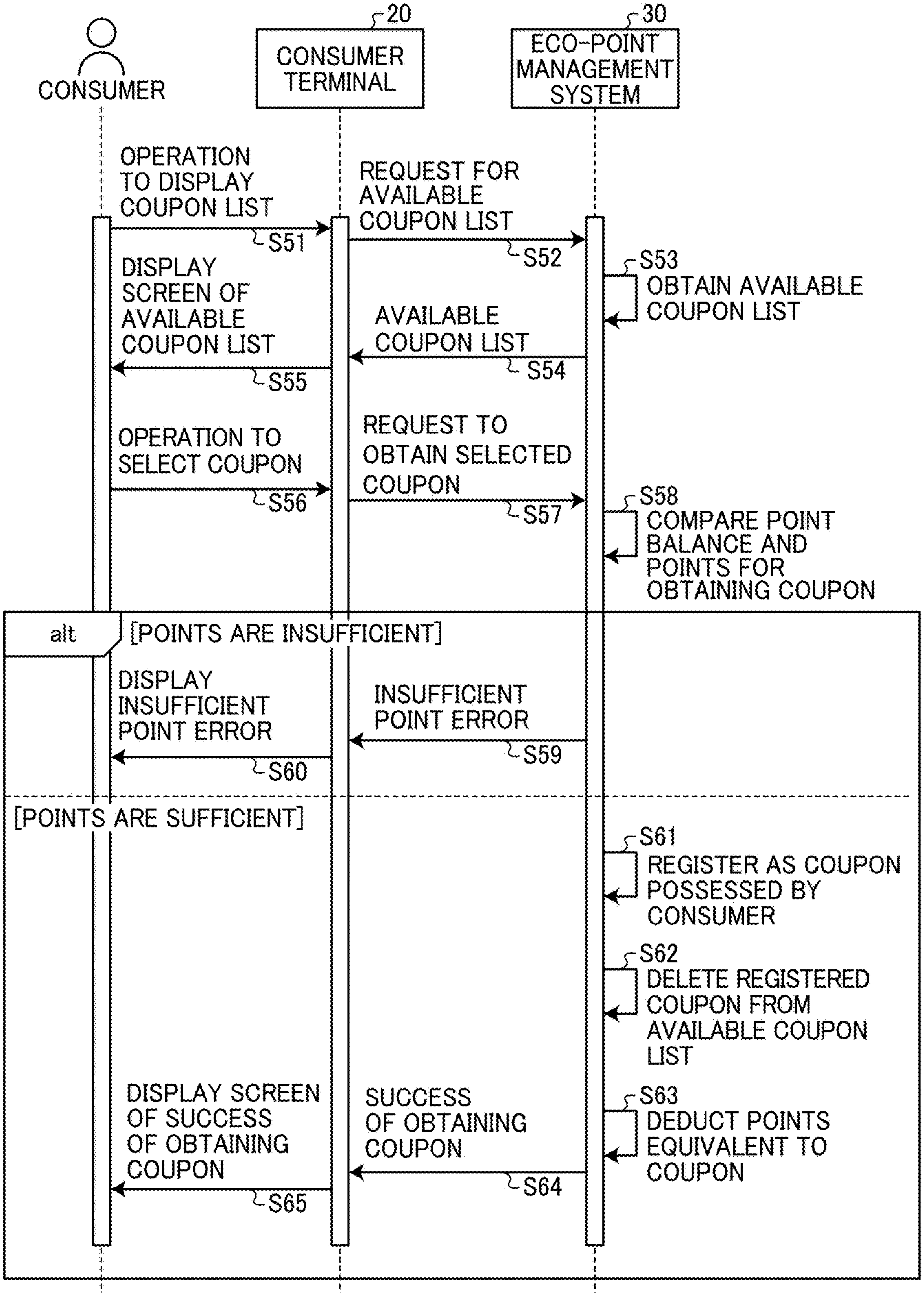
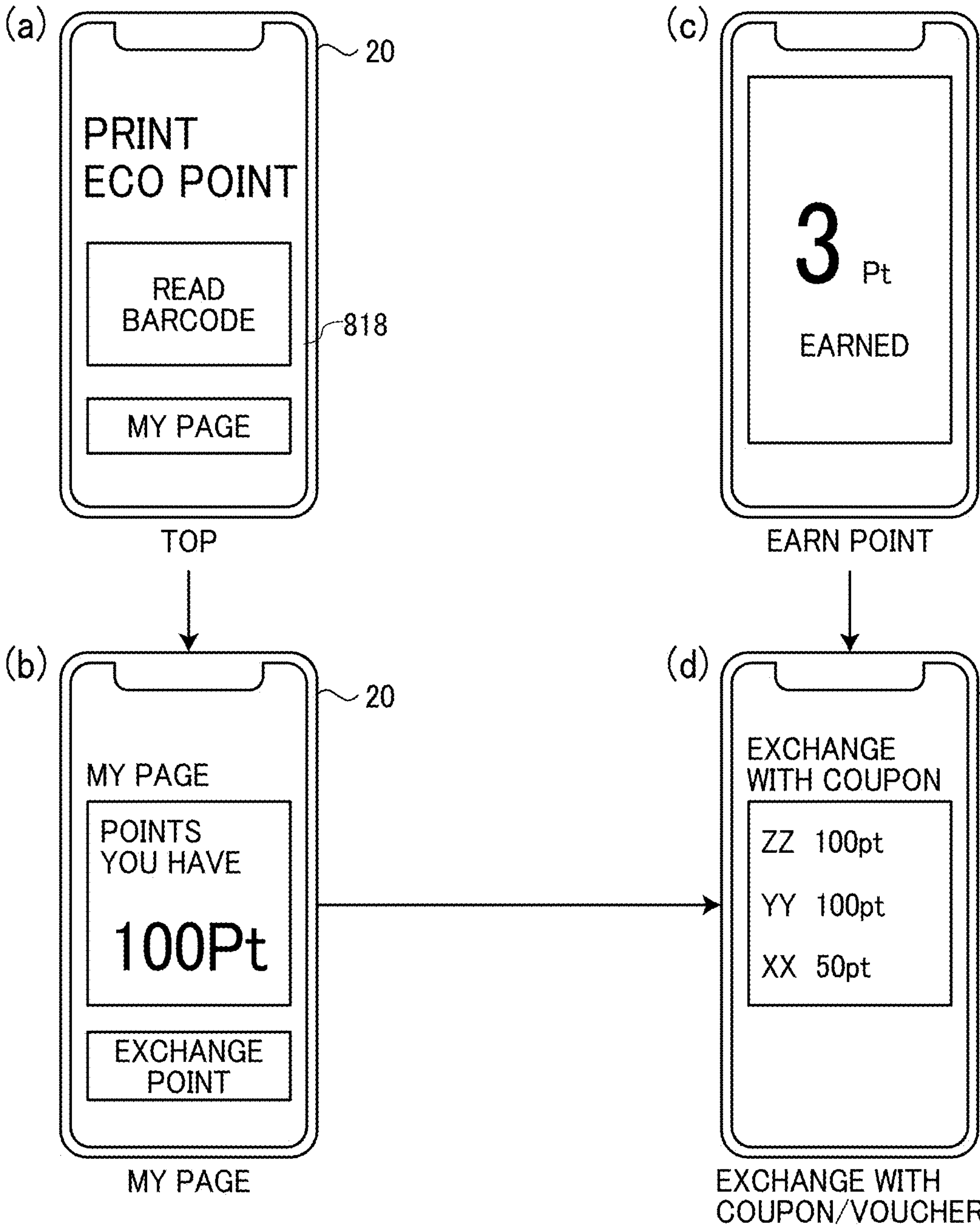

(a)
20
818
PRINT
ECO POINT
ID
Pass
LOGIN
LOGIN
(b)
20
PRINT
ECO POINT
READ
2D CODE
MY PAGE
TOP
(c)
20
READ QR CODE
(d)
20
3 Pt
EARNED
POINT

FIG. 14



CONSUMER
20
CONSUMER TERMINAL
30
ECO-POINT MANAGEMENT SYSTEM
OPERATION TO DISPLAY COUPON LIST
S51
REQUEST FOR AVAILABLE COUPON LIST
S52
S53
OBTAIN AVAILABLE COUPON LIST
DISPLAY SCREEN OF AVAILABLE COUPON LIST
AVAILABLE COUPON LIST
S54
S55
OPERATION TO SELECT COUPON
S56
REQUEST TO OBTAIN SELECTED COUPON
S57
S58
COMPARE POINT BALANCE AND POINTS FOR OBTAINING COUPON
alt
[POINTS ARE INSUFFICIENT]
DISPLAY INSUFFICIENT POINT ERROR
INSUFFICIENT POINT ERROR
S59
S60
[POINTS ARE SUFFICIENT]
S61
REGISTER AS COUPON POSSESSED BY CONSUMER
S62
DELETE REGISTERED COUPON FROM AVAILABLE COUPON LIST
S63
DEDUCT POINTS EQUIVALENT TO COUPON
DISPLAY SCREEN OF SUCCESS OF OBTAINING COUPON
SUCCESS OF OBTAINING COUPON
S64
S65

FIG. 15


(a)
20
PRINT
ECO POINT
READ
BARCODE
818
MY PAGE
TOP
(b)
20
MY PAGE
POINTS
YOU HAVE
100Pt
EXCHANGE
POINT
MY PAGE
(c)
20
3 Pt
EARNED
EARN POINT
(d)
20
EXCHANGE
WITH COUPON
ZZ 100pt
YY 100pt
XX 50pt
EXCHANGE WITH
COUPON/VOUCHER

POINT MANAGEMENT SYSTEM, POINT MANAGEMENT METHOD, AND NON-TRANSITORY RECORDING MEDIUM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This patent application is based on and claims priority pursuant to 35 U.S.C. § 119 (a) to Japanese Patent Application Nos. 2024-028746, filed on Feb. 28, 2024, and 2024-197301, filed on Nov. 12, 2024, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a point management system, a point management method, and a non-transitory recording medium.

Related Art

[0003] The butterfly mark of Japan Waterless Printing Association (WPA) is given to printed materials produced by environmentally friendly waterless printing that does not use moistening water.

[0004] On the carbon offset printed material produced by the members of Japan WPA, carbon dioxide (CO2) emissions in grams per copy calculated by Printing Goes Green, which is CO2 emission calculation software provided by Japan WPA, is indicated.

[0005] Further, there are techniques for measuring the consumption of printer resources and trading environmental offsets. For example, an automatic computer control system known in the art includes a printer resource search system for collecting data on consumed resources from a printer and controlling orders of environmental offset projects.

SUMMARY

[0006] The present disclosure described herein provides a point management system that includes circuitry to issue first identification information identifying a point for a consumer to obtain a coupon, register the first identification information in first association information, and calculate an actual value of a carbon dioxide emission from an amount of activity for a printed material. The printed material bears a symbol including the first identification information. The circuitry calculates the number of points based on the actual value, and registers the calculated points in the first association information in association with the first identification information.

[0007] The present disclosure described herein provides a point management method that includes issuing first identification information identifying a point for a consumer to obtain a coupon, registering the first identification information in first association information, and calculating an actual value of a carbon dioxide emission from an amount of activity for a printed material. The printed material bears a symbol including the first identification information. The method further includes calculating the number of points based on the actual value, and registering the calculated number of points in the first association information in association with the first identification information.

[0008] The present disclosure described herein provides a non-transitory recording medium storing a plurality of program codes which, when executed by one or more processors, causes the one or more processors to perform a method that includes issuing first identification information identifying a point for a consumer to obtain a coupon, registering the first identification information in first association information, and calculating an actual value of a carbon dioxide emission from an amount of activity for a printed material. The printed material bears a symbol including the first identification information. The method further includes calculating the number of points based on the actual value, and registering the calculated number of points in the first association information in association with the first identification information.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

[0010] FIG. **1** is a diagram illustrating an overall configuration of an information processing system;

[0011] FIG. **2** is a diagram illustrating an outline of an operation performed by the information processing system illustrated in FIG. **1**;

[0012] FIG. **3** is a diagram illustrating parties involved in an eco-point management system included in the information processing system illustrated in FIG. **1** and an outline of operations thereof;

[0013] FIG. **4** is a block diagram illustrating a hardware configuration of the eco-point management system illustrated in FIG. **3**;

[0014] FIG. **5** is a block diagram illustrating a hardware configuration of a consumer terminal;

[0015] FIG. **6** is a block diagram illustrating a hardware configuration of a printer;

[0016] FIG. **7** is a diagram illustrating a functional configuration of the information processing system illustrated in FIG. **1**;

[0017] FIG. **8** is a diagram illustrating a structure of data processed by the information processing system illustrated in FIG. **7**;

[0018] FIG. **9** is a sequence chart of a point registration process performed by the information processing system illustrated in FIG. **7**;

[0019] FIG. **10** is a diagram illustrating changes in a point management database (DB) in the point registration process illustrated in FIG. **9**;

[0020] FIG. **11** is a flowchart of a point obtaining process performed by the information processing system illustrated in FIG. **7**;

[0021] FIG. **12** is a diagram illustrating changes in the point management DB in the point obtaining process illustrated in FIG. **11**;

[0022] FIG. **13** is a diagram illustrating changes in the screen display on the consumer terminal in the point obtaining process illustrated in FIG. **11**;

[0023] FIG. **14** is a flowchart of a coupon exchange process performed by the information processing system illustrated in FIG. **7**; and

[0024] FIG. **15** is a diagram illustrating changes in the screen display of the consumer terminal in the coupon exchange process illustrated in FIG. **14**.

[0025] The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

[0026] In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.

[0027] Referring now to the drawings, a point management system, a point management method, and a program for point management according to embodiments of the present disclosure are described below with reference to the accompanying drawings. As used herein, the singular forms "a," "an," and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0028] The present invention, however, is not limited to the following embodiments, and constituent elements of the following embodiments include elements conceivable by those skilled in the art, substantially the same elements, and elements within so-called equivalent ranges. Furthermore, various omissions, substitutions, changes, and combinations of the constituent elements may be made without departing from the gist of the following embodiments.

Overall Configuration and Operation of Information Processing System

[0029] FIG. **1** is a diagram illustrating an overall configuration of an information processing system **1**. FIG. **2** is a diagram illustrating an outline of an operation performed by the information processing system **1**. FIG. **3** is a diagram illustrating parties involved in an eco-point management system **30** of the information processing system **1** and an outline of operations performed by the eco-point management system **30**. A description is given of the overall configuration of the information processing system **1** with reference to FIGS. **1** to **3**.

[0030] The information processing system **1** illustrated in FIG. **1** registers an eco-point based on the actual carbon dioxide emission generated in producing a printed material and enables a consumer of the product to use the eco-point. Eco-points are points given to a consumer who purchases an environmentally friendly product or service, and eco-points provide some benefit to the consumer when used. The eco-points may be referred to simply as points in the following description. As illustrated in FIG. **1**, the information processing system **1** includes a brand owner terminal **10** (second information processing terminal), a consumer terminal **20** (first information processing terminal), an eco-point management system **30**, a printing management system **40**, a printing company terminal **41**, and a printer **42**. The printing management system **40**, the printing company terminal **41**, and the printer **42** are owned by a printing company. The brand owner terminal **10**, the consumer terminal **20**, the eco-point management system **30**, and the printing management system **40** perform data communication via a network N. The printing management system **40**, the printing company terminal **41**, and the printer **42** owned by the printing company are connected to a local area network (LAN) to perform data communication.

[0031] The brand owner terminal **10** is an information processing terminal used by a brand owner. The brand owner is a company that receives a service provided by the eco-point management system **30**, sells a printed material (product) bearing a symbol such as a QUICK RESPONSE (QR) CODE to consumers, and posts, on an app, coupons, vouchers, or both of coupons and vouchers (may be collectively referred to as coupons in the following description) for a product of the company, thereby encouraging the consumers to purchase new products. A coupon is a virtual ticket authorizing a consumer to purchase or exchange a product or service at a discount price and is a concept including the above-described voucher. The brand owner corresponds to a "provider." As illustrated in FIGS. **2** and **3**, the brand owner terminal **10** requests the eco-point management system **30** to register a coupon for a product of the brand owner. Then, the eco-point management system **30** registers the information on the coupon in a coupon-voucher management database (DB) **310***b*.

[0032] The consumer terminal **20** is an information processing terminal such as a personal computer (PC) used by a consumer who purchases a product provided by the brand owner. As illustrated in FIGS. **2** and **3**, the consumer terminal **20** reads a two-dimensional code (QR CODE in the present embodiment) printed on a product purchased by a consumer in response to an operation of the consumer, and requests the eco-point management system **30** to obtain a point associated with the point identifier (ID) included in the QR CODE. The point ID is first identification information. The eco-point management system **30** adds the number of points associated with the point ID to the points owned by the consumer in association with the consumer in a point management DB **310***c*. The point management DB **310***c* is first association information. As illustrated in FIGS. **2** and **3**, in response to an operation of the consumer, the consumer terminal **20** transmits to the eco-point management system **30** a request to obtain a coupon in exchange for the point earned (obtained) by the consumer. Then, the eco-point management system **30** registers the coupon as a coupon obtained by the consumer. Then, the consumer can use the obtained coupon via the consumer terminal **20**.

[0033] Although the QR CODE is printed on the product in the description above, any symbol in which information is encoded or included may be used. Examples of the symbol include other two-dimensional codes such as Portable Data File (PDF) **417** or DataMatrix, binary codes, and one-dimensional codes such as barcodes. Alternatively, the symbol may be the combination of any of these codes. In the following description, the symbol in which a point ID is included or encoded is a QR CODE.

[0034] The eco-point management system **30** is an information processing apparatus or an information processing system that registers, in response to a request from a printing company, a point corresponding to the actual carbon dioxide emission generated in producing a printed material, and provides the service of allowing the consumer who purchases the printed material (product) to earn the point and exchange the point with a coupon registered by the brand owner.

[0035] The printing management system **40** is implemented by a server apparatus or a server system that manages execution of a printing process. As illustrated in FIGS. **2** and **3**, the printing management system **40** requests, in response to an instruction from the printing company, the eco-point management system **30** to register a point ID, and requests the registration of the point corresponding to the actual carbon dioxide emission generated in the printing of the printed material by the printer **42**. The eco-point management system **30** calculates the number of points by using the actual carbon dioxide emission and the emission intensity stored in a CO2 emission calculation DB **310***d*. The emission intensity indicates the carbon dioxide emission per unit of activity amount. The eco-point management system **30** registers the calculated number of points in the coupon-voucher management DB **310***b*. The activity amount indicates, for each process, the amount of activity or processing that generates carbon dioxide emission in printing. Examples of the activity amount include the electricity consumed by the printer **42**, the toner and the number of sheets consumed in the printing, and the number of sheets discarded as waste sheets.

[0036] The printing company terminal **41** is an information processing terminal such as a PC for instructing execution of a job.

[0037] The printer **42** is an image forming apparatus that executes a job to perform printing in response to an instruction from the printing management system **40**.

Hardware Configuration of Eco-Point Management System

[0038] FIG. **4** is a block diagram illustrating a hardware configuration of the eco-point management system **30**. A description is given below of the hardware configuration of the eco-point management system **30** with reference to FIG. **4**.

[0039] As illustrated in FIG. **4**, the eco-point management system **30** includes a central processing unit (CPU) **701**, a read-only memory (ROM) **702**, a random-access memory (RAM) **703**, an auxiliary memory **705**, a medium drive **707**, a display **708**, a network interface (I/F) **709**, a keyboard **711**, a mouse **712**, and a digital versatile disk (DVD) drive **714**.

[0040] The CPU **701** is a processor that controls the entire operation of the eco-point management system **30**. The ROM **702** is a nonvolatile memory that stores a program for the eco-point management system **30**. The RAM **703** is a volatile memory used as a work area for the CPU **701**.

[0041] The auxiliary memory **705** is a memory such as a hard disk drive (HDD) or a solid-state drive (SSD) that stores various data or programs. The medium drive **707** controls the reading and writing of data from and to a storage medium **706** such as a flash memory under the control of the CPU **701**.

[0042] The display **708** is implemented by, for example, a liquid crystal display (LCD) or an organic electro-luminescence (EL) display, which displays various types of information such as a cursor, a menu, a window, characters, and an image.

[0043] The network I/F **709** is an interface circuit that performs data communication with an external device such as the brand owner terminal **10**, the consumer terminal **20**, and the printing management system **40** via the network N. The network I/F **709** is, for example, a network interface card (NIC) compliant with ETHERNET and can establish wired or wireless communications in compliance with Transmission Control Protocol (TCP)/Internet protocol (IP).

[0044] The keyboard **711** is an input device used for inputting operations such as selecting characters, numerals, or various instructions, and for moving a cursor. The mouse **712** is an input device used for inputting operations such as selecting and executing various instructions, selecting an object to be processed, and moving a cursor.

[0045] The DVD drive **714** is a device that reads or writes data to or from a DVD **713** such as a DVD-ROM or a digital versatile disk recordable (DVD-R), which is an example of a removable storage medium.

[0046] The CPU **701**, the ROM **702**, the RAM **703**, the auxiliary memory **705**, the medium drive **707**, the display **708**, the network I/F **709**, the keyboard **711**, the mouse **712**, and the DVD drive **714** are connected to communicate through a bus line **710** such as an address bus or a data bus.

[0047] The hardware configuration of the eco-point management system **30** is not limited to that illustrated in FIG. **4**. The eco-point management system **30** does not necessarily include all the components illustrated in FIG. **4** or may include other components. The eco-point management system **30** is not necessarily implemented by a single information processing apparatus as illustrated in FIG. **4** but may be implemented by multiple information processing apparatuses that communicate via a network.

[0048] The hardware configuration of the printing management system **40** is similar to the hardware configuration illustrated in FIG. **4**.

Hardware Configuration of Consumer Terminal

[0049] FIG. **5** is a block diagram illustrating a hardware configuration of the consumer terminal **20**. The hardware configuration of the consumer terminal **20** is described below with reference to FIG. **5**.

[0050] As illustrated in FIG. **5**, the consumer terminal **20** includes a CPU **801**, a ROM **802**, a RAM **803**, an electrically erasable programmable read-only memory (EEPROM) **804**, an imaging device **805**, an imaging I/F **806**, an acceleration and direction sensor **807**, and a global navigation satellite system (GNSS) receiver **808**.

[0051] The CPU **801** is a processor that controls the entire operation of the consumer terminal **20**. The ROM **802** is a non-volatile memory that stores a program, such as an initial program loader (IPL), used for driving the CPU **801**. The RAM **803** is a volatile memory used as a work area for the CPU **801**. The EEPROM **804** is a non-volatile memory that stores various data such as a control program.

[0052] The imaging device **805** is a built-in imaging device (e.g., a camera) that captures an image of a subject to obtain image data under the control of the CPU **801**. The imaging device **805** employs an image sensor such as a complementary metal oxide semiconductor (CMOS). In alternative to the CMOS image sensor, an imaging element such as a charge-coupled device (CCD) sensor may be used. The imaging I/F **806**, which may be implemented by an interface circuit, is an interface used to control the driving of the imaging device **805**.

[0053] The acceleration and direction sensor **807** includes various sensors such as an electromagnetic compass to detect geomagnetism, a gyrocompass, and an acceleration sensor.

[0054] The GNSS receiver **808** is a receiver that receives positioning signals from positioning satellites. The GNSS receiver **808** receives, for example, a global positioning system (GPS) signal from a GPS satellite.

[0055] As illustrated in FIG. **5**, the consumer terminal **20** further includes a long-range communication circuit **810**, an antenna **810***a*, a short-range communication circuit **811**, an antenna **811***a*, a microphone **812**, a speaker **813**, an audio input and output (I/O) I/F **814**, a display **815**, an external device connection I/F **816**, a vibrator **817**, and a touch panel **818**.

[0056] The long-range communication circuit **810** is a communication circuit that performs wireless communication with other devices via the antenna **810***a* in compliance with a protocol such as Wireless Fidelity (WI-FI) through the network N.

[0057] The short-range communication circuit **811** is a communication circuit that performs short-range wireless communication with another device via the antenna **811***a* in compliance with a communication protocol such as near field communication (NFC) or BLUETOOTH.

[0058] The microphone **812** is a built-in sound collecting circuit that converts collected sound into an electrical signal. The speaker **813** is a built-in circuit that converts the electrical signal into physical vibration to output sounds such as music or voice.

[0059] The audio I/O I/F **814** is an interface circuit that inputs and outputs audio signals between the microphone **812** and the speaker **813** under the control of the CPU **801**. The microphone **812** and the speaker **813** may be parts of a wireless headset.

[0060] The display **815** is a display device such as a liquid crystal display or an organic EL display that displays, for example, an image of an object and various icons. The external device connection I/F **816** is an interface circuit that connects to various external devices in compliance with a communication protocol such as the Universal Serial Bus (USB).

[0061] The vibrator **817** is a device that generates physical vibration under the control of the CPU **801**.

[0062] The touch panel **818** is an input device that receives a user's touch operation on the screen of the display **815** to cause the consumer terminal **20** to provide various functions.

[0063] The CPU **801**, the ROM **802**, the RAM **803**, the EEPROM **804**, the imaging I/F **806**, the acceleration and direction sensor **807**, the GNSS receiver **808**, the long-range communication circuit **810**, the short-range communication circuit **811**, the audio I/O I/F **814**, the display **815**, the external device connection I/F **816**, the vibrator **817**, and the touch panel **818** are connected to communicate with each other via a bus line **809** such as an address bus or a data bus.

[0064] Note that the hardware configuration of the consumer terminal **20** illustrated in FIG. **5** is one example. Some of the illustrated components can be omitted, or one or more different devices may be added.

[0065] Hardware Configuration of Printer FIG. **6** is a block diagram illustrating a hardware configuration of the printer **42**. The hardware configuration of the printer **42** is described below with reference to FIG. **6**.

[0066] As illustrated in FIG. **6**, the printer **42** includes a controller **910**, a short-range communication circuit **920**, an engine controller **930**, a control panel **940**, and a network I/F **950**.

[0067] The controller **910** includes a CPU **901** as a main processor, a system memory **902**, a northbridge (NB) **903**, a southbridge (SB) **904**, an application-specific integrated circuit (ASIC) **906**, a local memory **907**, a hard disk drive (HDD) controller **908**, and a hard disk (HD) **909**. The NB **903** and the ASIC **906** are connected by an accelerated graphics port (AGP) bus **921**.

[0068] The CPU **901** is an arithmetic logic device that controls the entire operation of the printer **42**. The NB **903** connects the CPU **901** to the system memory **902**, the SB **904**, and the AGP bus **921**. The NB **903** includes a memory controller that controls the reading or writing of various data from or to the system memory **902**, a peripheral component interconnect (PCI) master, and an AGP target.

[0069] The system memory **902** includes a ROM **902***a* that stores programs and data for implementing various functions of the controller **910** and a RAM **902***b* used as a memory for deploying programs or data or for loading drawing data in printing. The programs stored in the RAM **902***b* may be stored, as a file installable or executable by the computer, in a computer-readable storage medium, such as a compact disc-read-only memory (CD-ROM), compact disc-recordable (CD-R), or digital versatile disc (DVD).

[0070] The SB **904** is a bridge that connects the NB **903** to, for example, a PCI device or a peripheral device. The ASIC **906** is an integrated circuit (IC) for image processing having a hardware element for image processing and acts as a bridge that connects the AGP bus **921**, a PCI bus **922**, the HDD controller **908**, and the local memory **907** to each other. The ASIC **906** includes a PCI target, an AGP master, an arbiter (ARB) as a core element of the ASIC **906**, a memory controller that controls the local memory **907**, multiple Direct Memory Access Controllers (DMACs) that convert coordinates of image data with a hardware logic, and a PCI unit that transfers data between a scanner controller **931** and a printer controller **932** through the PCI bus **922**. The ASIC **906** may be connected to a USB interface or an Institute of Electrical and Electronics Engineers 1394 (IEEE1394) interface.

[0071] The local memory **907** is used as a buffer for an image to be copied or a buffer for coding. The HD **909** is a storage for storing image data, font data used in printing, and forms. The HDD controller **908** controls the reading or writing of various data to or from the HD **909** under the control of the CPU **901**. The HDD controller **908** and the HD **909** may be replaced by an SSD.

[0072] The AGP bus **921** is a bus interface for a graphics accelerator card proposed to accelerate graphics processing. The AGP bus **921** increases the speed of the graphics accelerator card by directly accessing the system memory **902** with high throughput.

[0073] The short-range communication circuit **920** is a communication circuit in compliance with a communication protocol such as NFC or BLUETOOTH. The short-range communication circuit **920** is electrically connected to the ASIC **906** through the PCI bus **922**. The short-range communication circuit **920** is provided with an antenna **920***a* for wireless communication.

[0074] The engine controller **930** includes the scanner controller **931** and the printer controller **932**. The scanner controller **931** and the printer controller **932** each have an image processing function such as error diffusion or gamma conversion.

[0075] The control panel **940** includes a panel display **940***a* such as a touch panel and a hard keypad **940***b*. The panel display **940***a* displays current settings or a selection screen to receive a user input. The hard keypad **940***b* includes a numeric keypad that receives setting values of various image forming parameters such as an image density parameter and a start key that receives an instruction for starting copying.

[0076] In response to an instruction to select a specific application input using a mode switch key on the control panel **940**, the printer **42** sequentially selects a document server function, a copy function, a print function, and a facsimile communication function. When the document server function is selected, the printer **43** operates in a document server mode. When the copier function is selected, the printer **43** operates in a copy mode. When the printer function is selected, the printer **43** operates in a print mode. When the facsimile communication function is selected, the printer **43** operates in a facsimile communication mode.

[0077] The network I/F **950** is an interface circuit that perform data transmission via a network, in compliance with a communication protocol such as ETHERNET or TCP/IP. The network I/F **950** is electrically connected to the ASIC **906** through the PCI bus **922**.

[0078] The hardware configuration of the printer **42** is not limited to that illustrated in FIG. **6**. The printer **42** does not necessarily include all the components illustrated in FIG. **6** or may include other components.

[0079] Functional Configuration and Operation of Information Processing System FIG. **7** is a diagram illustrating a functional configuration of the information processing system **1**. FIG. **8** is a diagram illustrating a structure of the data processed by the information processing system **1**. The functional configuration and the operation of the information processing system **1** are described with reference to FIGS. **7** and **8**.

[0080] As illustrated in FIG. **7**, the consumer terminal **20** includes a communication unit **201**, a reading control unit **202**, a point request unit **203**, a coupon list request unit **204**, a coupon request unit **205**, and a display control unit **206**.

[0081] The communication unit **201** is a functional unit that performs data communication with other devices via the long-range communication circuit **810**.

[0082] The reading control unit **202** is a functional unit that, in response to an operation on the touch panel **818** by the consumer, controls the imaging device **805** to read a QR CODE printed on a printed material (product) and decodes the read QR CODE to obtain a point ID which is identification information of a point.

[0083] The point request unit **203** is a functional unit that transmits a request to obtain the points associated with the point ID included in the QR CODE obtained by the reading control unit **202** to the eco-point management system **30** via the communication unit **201**.

[0084] The coupon list request unit **204** is a functional unit that, in response to an operation on the touch panel **818** by the consumer, transmits a request to obtain a list of coupons available to the consumer to the eco-point management system **30** via the communication unit **201**.

[0085] The coupon request unit **205** is a functional unit that transmits a request to obtain a coupon selected by the consumer from the list of available coupons to the eco-point management system **30** via the communication unit **201**.

[0086] The display control unit **206** is a functional unit that controls the display operation performed by the display **815**.

[0087] The communication unit **201**, the reading control unit **202**, the point request unit **203**, the coupon list request unit **204**, the coupon request unit **205**, and the display control unit **206** are implemented by the CPU **801** executing an application **200** illustrated in FIG. **5**. The functions of the communication unit **201**, the reading control unit **202**, the point request unit **203**, the coupon list request unit **204**, the coupon request unit **205**, and the display control unit **206** may be partly or entirely implemented by a hardware circuit (integrated circuit) such as a field-programmable gate array (FPGA) or an application specific integrated circuit (ASIC), in alternative to the CPU **801** that executes the software program.

[0088] The functionals unit of the consumer terminal **20** illustrated in FIG. **7** are each a conceptual representation of the function, and the functional configuration of the consumer terminal **20** is not limited thereto. For example, the multiple functional units of the consumer terminal **20** illustrated as independent units in FIG. **7** may be configured as a single functional unit. The functions provided by a single functional unit of the consumer terminal **20** illustrated in FIG. **7** may be divided and allocated to multiple functional units. Further, the functional units of the consumer terminal **20** are not necessarily software modules clearly configured as individual blocks as illustrated in FIG. **7**, and the functions of the functional units may be implemented as a whole by the execution of the application **200** on the consumer terminal **20**.

[0089] As illustrated in FIG. **7**, the eco-point management system **30** includes a communication unit **301** (a reception unit), a coupon registration unit **302** (a third registration unit), an ID registration unit **303** (a first registration unit), a point calculation unit **304** (calculation unit), a point registration unit **305** (second registration unit), a determination unit **306**, a point addition unit **307** (addition unit), a coupon list obtaining unit **308**, a coupon processing unit **309** (processing unit), and a storage unit **310**.

[0090] The communication unit **301** is a functional unit that performs data transmission with other devices via the network I/F **709**.

[0091] The coupon registration unit **302** is a functional unit that, when the communication unit **301** receives a registration request to register a coupon from the brand owner terminals **10**, registers information on the coupon in the coupon-voucher management DB **310***b* in the storage unit **310**. The coupon-voucher management DB **310***b* (second association information) is a database in, for example, a table format for managing pieces of information relating to a coupon in association with each other. As illustrated in FIG. **8**, the coupon-voucher management DB **310***b* stores the information (e.g., identification information) on the brand owner (third identification information), the name of the product to which the coupon is applied, the number of points for obtaining the coupon, the store where the coupon is usable, the number of issued coupons, the expiration date of the coupon, and conditions for obtaining the coupon.

[0092] The ID registration unit **303** is a functional unit that, when the communication unit **301** receives a registration request for a point ID from the printing management system **40**, newly issues a point ID and registers the point ID in the point management DB **310***c* (first association information) in the storage unit **310**. The point management DB **310***c* is a database in, for example, a table format for managing pieces of information relating to a point in association with each other. As illustrated in FIG. **8**, the point management DB **310***c* stores the point ID, the brand owner who provides the product assigned with the points associated with the point ID, the number of points associated with the point ID, and the consumer ID (second identification information) identifying the consumer who have earned the point. The consumer ID may be referred to as a user ID in the following description. The ID registration unit **303** first registers the point ID and the brand owner corresponding to the point ID in association with each other when registering the point ID in the point management DB **310***c*.

[0093] When the point calculation unit **304** is a functional unit that, when the communication unit **301** receives from the printing management system **40** a registration request for a point, calculates a predicted value and an actual value of carbon dioxide emission from the amount of printing activity received together with the registration request for the point and the emission intensity indicating the carbon dioxide emission per unit of activity amount, stored in the CO2 emission calculation DB **310***d*, calculates the number of points from the predicted value and the actual value, and registers the calculated number of points in the coupon-voucher management DB **310***b*.

[0094] For example, the point calculation unit **304** obtains the predicted value of carbon dioxide emission using a learned model that calculates a predicted value of the carbon dioxide emission from the amount of activity of printing, and calculates the actual carbon dioxide emission from the product of the amount of printing activity and the emission intensity obtained from the CO2 emission calculation DB **310***d*. Then, the point calculation unit **304** calculates the number of points by Equation 1 below.

$$\text{Point} = \text{Predicted CO2 emission} - \text{Actual CO2 emission amount} \quad (1)$$

[0095] In other words, the smaller the actual value compared with the predicted value (the larger the carbon dioxide reduction), the greater the number of points calculated. The CO2 emission calculation DB **310***d* is a database for managing information such as the emission intensity of the activity amount as illustrated in FIG. **8**. The information stored in the CO2 emission calculation DB **310***d* is used to calculate carbon dioxide emission.

[0096] Alternative to calculating the number of points using the above Equation 1, the point calculation unit **304** may calculate the number of points by multiplying the value of Equation 1 by a coefficient. Yet alternatively, the point calculation unit **304** may calculate the number of points by adding a numerical value indicating, for example, the number of log-in days in a campaign period to the right side of Equation 1, or may add some point to the right side of Equation 1 to encourage the consumer to purchase the product as a limited-time promotion. The point calculation unit **304** may change the above-described point calculation method for, for example, on an individual industry basis or an individual manufacturer basis.

[0097] The point registration unit **305** is a functional unit that registers the number of points calculated by the point calculation unit **304** in association with the point ID registered in the point management DB **310***c* by the ID registration unit **303**.

[0098] The determination unit **306** is a functional unit that, when the communication unit **301** receives the point request from the consumer terminal **20**, determines whether one or more points associated with the point ID received together with the point request has already been given. To be specific, the determination unit **306** refers to the point management DB **310***c* and determines whether the point has already been given depending on whether a user ID associated with the point ID has already been registered. That is, when the user ID associated with the point ID is already registered, the determination unit **306** determines that the point associated with the point ID has already been given.

[0099] The point addition unit **307** is a functional unit that, when the determination unit **306** determines that the point associated with the point ID is not an already-given point, refers to a user management DB **310***a* in the storage unit **310** and adds the point associated with the point ID to the point balance associated with to the user ID identifying the consumer who uses the consumer terminal **20** that has transmitted the point request received by the communication unit **301**. The user management DB **310***a* (management information) is a database in, for example, the table format for managing the user ID, a password for login, and the point balance indicating the points earned by the users in association with each other as illustrated in FIG. **8**.

[0100] The coupon list obtaining unit **308** is a functional unit that, when the communication unit **301** receives a request to obtain a list of coupons from the consumer terminal **20**, refers to the coupon-voucher management DB **310***b* and obtains a list of coupons available to the consumer who uses the consumer terminal **20**. For example, the coupon list obtaining unit **308** obtains a list of coupons that satisfy the expiration date or the conditions among the coupons registered in the coupon-voucher management DB **310***b*.

[0101] The coupon processing unit **309** is a functional unit that, when the communication unit **301** receives a request to obtain a coupon selected by a consumer from the consumer terminal **20**, refers to the user management DB **310***a* and sets the coupon to be a possessed coupon by the consumer when the number of points for obtaining the coupon is equal to or less than the number of points possessed by the consumer.

[0102] The storage unit **310** is a functional unit that stores the user management DB **310***a*, the coupon-voucher management DB **310***b*, the point management DB **310***c*, the CO2 emission calculation DB **310***d* illustrated in FIG. **8**, etc. The storage unit **310** is implemented by the auxiliary memory **705** illustrated in FIG. **4**.

[0103] The communication unit **301**, the coupon registration unit **302**, the ID registration unit **303**, the point calculation unit **304**, the point registration unit **305**, the determination unit **306**, the point addition unit **307**, the coupon list obtaining unit **308**, and the coupon processing unit **309** are implemented by the CPU **701** illustrated in FIG. **4** executing a program. A part or all of the communication unit **301**, the coupon registration unit **302**, the ID registration unit **303**, the point calculation unit **304**, the point registration unit **305**, the determination unit **306**, the point addition unit **307**, the coupon list obtaining unit **308**, and the coupon processing unit **309** may be implemented by a hardware circuit (integrated circuit) such as an FPGA or an ASIC instead of the CPU **701** executing the software program.

[0104] The functional units of the eco-point management system **30** illustrated in FIG. **7** are conceptual representations of the functions and may be implemented in various other ways. For example, the multiple functional units of the eco-point management system **30** illustrated as independent functional units in FIG. **7** may be configured as a single functional unit. Alternatively, the function of one functional unit of the eco-point management system **30** illustrated in FIG. **7** may be divided into multiple functions and allocated to multiple functional units. Further, the functional units of the eco-point management system **30** are not necessarily software modules clearly configured as individual blocks as illustrated in FIG. **7**, and the functions of the functional units may be implemented as a whole by the execution of a program on the eco-point management system **30**.

[0105] As illustrated in FIG. **7**, the printing management system **40** includes a communication unit **401**, an ID registration request unit **402**, a code generation unit **403**, a printing control unit **404**, and a point registration request unit **405**.

[0106] The communication unit **401** is a functional unit that performs data transmission with other devices via the network I/F **709**.

[0107] The ID registration request unit **402** is a functional unit that, when the communication unit **401** receives a job execution command for printing from the printing company terminal **41**, transmits a point ID registration request for the point corresponding to the printed material created in the job to the eco-point management system **30** via the communication unit **401**.

[0108] The code generation unit **403** is a functional unit that, when the communication unit **401** receives the point ID from the eco-point management system **30**, generates a QR CODE including the information on the point ID. The QR CODE may include, for example, a uniform resource locator (URL) for accessing the eco-point management system **30** when the QR CODE is read.

[0109] The printing control unit **404** is a functional unit that controls the operation of the printer **42**. For example, the printing control unit **404** prints the QR CODE generated by the code generation unit **403** on a printed material (product).

[0110] The point registration request unit **405** is a functional unit that calculates the activity amount of the job instructed by the job execution command received by the communication unit **401** and transmits a point registration request in addition to the activity amount to the eco-point management system **30** via the communication unit **401**.

[0111] The communication unit **401**, the ID registration request unit **402**, the code generation unit **403**, the printing control unit **404**, and the point registration request unit **405** are implemented by the CPU **701** illustrated in FIG. **4** executing a program. The functions of the communication unit **401**, the ID registration request unit **402**, the code generation unit **403**, the printing control unit **404**, and the point registration request unit **405** may be partly or entirely implemented by a hardware circuit (integrated circuit) such as an FPGA or an ASIC instead of the CPU **701** executing the software program.

[0112] The functional units of the printing management system **40** illustrated in FIG. **7** are conceptual representations of the functions, and the functional configuration thereof is not limited thereto. For example, the multiple functional units of the printing management system **40** illustrated as independent functional units in FIG. **7** may be configured as one functional unit. Alternatively, the function performed by one functional unit of the printing management system **40** illustrated in FIG. **7** may be divided and allocated to multiple functional units. Further, the functional units of the printing management system **40** are not necessarily software modules clearly configured as individual blocks as illustrated in FIG. **7** and the functions of the functional units may be implemented as a whole by the execution of a program on the printing management system **40**.

Point Registration Process by Information Processing System

[0113] FIG. **9** is a sequence chart of a point registration process performed by the information processing system **1**. FIG. **10** is a diagram illustrating changes in the point management DB **310***c* in the point registration process performed by the information processing system **1**. The point registration process performed by the information processing system **1** is described with reference to FIGS. **9** and **10**.

Step S**11**

[0114] The printing company terminal **41** transmits a job execution command instructing printing to create a desired product to the printing management system **40** in response to an operation by printing company staff. The communication unit **401** of the printing management system **40** receives the job execution command.

Step S**12**

[0115] When the communication unit **401** receives the job execution command for printing from the printing company terminal **41**, the ID registration request unit **402** of the printing management system **40** transmits a point ID registration request for the point corresponding to the printed material created in the job to the eco-point management system **30** via the communication unit **401**. The communication unit **301** of the eco-point management system **30** receives the point ID registration request.

Step S**13**

[0116] When the communication unit **301** receives the point ID registration request from the printing management system **40**, the ID registration unit **303** of the eco-point management system **30** newly issues a point ID and registers the point ID in the point management DB **310***c* in the storage unit **310**. The issued point ID is a unique ID that does not overlap with already issued point IDs. When the ID registration unit **303** registers the point ID in the point management DB **310***c*, the brand owner corresponding to the point ID is associated with the point ID as illustrated in FIG. **10**.

Step S**14**

[0117] The ID registration unit **303** transmits the issued point ID to the printing management system **40** via the communication unit **301**. The communication unit **401** of the printing management system **40** receives the point ID.

Step S15

[0118] When the communication unit **401** receives the point ID from the eco-point management system **30**, the code generation unit **403** of the printing management system **40** generates a two-dimensional code (e.g., QR CODE or PDF417) including the information on the point ID.

Step S16

[0119] The printing control unit **404** of the printing management system **40** prints the QR CODE generated by the code generation unit **403** on a printed material (product).

Step S17

[0120] The point registration request unit **405** of the printing management system **40** calculates the activity amount of the job instructed by the job execution command received by the communication unit **401**, and transmits the activity amount and the point registration request to the eco-point management system **30** via the communication unit **401**. Then, the communication unit **301** of the eco-point management system **30** receives the activity amount and the point registration request.

Step S18

[0121] When the communication unit **301** receives the activity amount and the point registration request from the printing management system **40**, the point calculation unit **304** of the eco-point management system **30** obtains the emission intensity from the CO2 emission calculation DB **310***d*.

Step S19

[0122] The point calculation unit **304** calculates the predicted value and the actual value of carbon dioxide emission from the activity amount and the emission intensity, calculates the points from the above-described Equation 1, and registers the points in the coupon-voucher management DB **310***b*.

[0123] Alternative to calculating the points using the above Equation 1, the point calculation unit **304** may calculate the points by multiplying the value of Equation 1 by a coefficient. Yet alternatively, the point calculation unit **304** may calculate the points by adding a numerical value indicating, for example, the number of log-in days in a campaign period to the right side of Equation 1, or may add some points to the right side of Equation 1 to encourage the consumer to purchase the product as a limited-time promotion. The point calculation unit **304** may change the above-described point calculation method for, for example, on an individual industry basis or an individual manufacturer basis.

Step S20

[0124] As illustrated in FIG. **10**, the point registration unit **305** of the eco-point management system **30** registers the number of points calculated by the point calculation unit **304** in association with the point ID registered in the point management DB **310***c* by the ID registration unit **303**.

Step S21

[0125] The point registration unit **305** transmits a notification indicating that the point registration has succeeded to the printing management system **40** via the communication unit **301**. Then, the communication unit **401** of the printing management system **40** receives the notification indicating that the point registration has succeeded.

Step S22

[0126] The communication unit **401** of the printing management system **40** transmits the notification indicating that the point registration has succeeded, which is received from the eco-point management system **30**, to the printing company terminal **41**. The printing company terminal **41** displays information that the point registration has succeeded. Then, the point registration process ends.

[0127] In the point registration process illustrated in FIG. **9**, the points are calculated and registered after the QR CODE is printed, but the sequence is not limited thereto. Alternatively, the points may be calculated and registered after the printed material is created, and the QR CODE may be printed on the printed material after the points are determined.

Process of Obtaining Point by Information Processing System

[0128] FIG. **11** is a flowchart of the point obtaining process performed by the information processing system **1**. FIG. **12** is a diagram illustrating changes in the point management DB **310***c* in the point obtaining process performed by the information processing system **1**. FIG. **13** is a diagram illustrating changes in the screen display on the consumer terminal **20** in the point obtaining process performed by the information processing system **1**. The point obtaining process performed by the information processing system **1** is described with reference to FIGS. **11** to **13**.

Step S31

[0129] In response to an operation on the touch panel **818** by the consumer, the reading control unit **202** of the consumer terminal **20** controls the imaging device **805** to read the two-dimensional code (e.g., QR CODE or PDF417) printed on the printed material (product), and decodes the QR CODE to obtain the point ID identifying the point. For example, the consumer performs a login operation on the screen of the consumer terminal **20** illustrated in (a) of FIG. **13**, and taps the button for reading a two-dimensional 2D code (e.g., QR CODE) on the top screen illustrated in (b) of FIG. **13**.

[0130] When the QR CODE is read by the reading control unit **202**, the display control unit **206** displays the read QR CODE on the display **815** as illustrated in (c) of FIG. **13**.

Step S32

[0131] The point request unit **203** of the consumer terminal **20** transmits a point request to obtain the point associated with the point ID included in the QR CODE obtained by the reading control unit **202** to the eco-point management system **30** via the communication unit **201**. The communication unit **301** of the eco-point management system **30** receives the point request and the point ID.

Step S33

[0132] When the communication unit **301** receives the point request from the consumer terminal **20**, the determination unit **306** of the eco-point management system **30** refers to the point management DB **310***c* and obtains (retrieves) the number of points associated with the point ID received together with the point request.

Step S34

[0133] The determination unit **306** refers to the point management DB **310***c* and determines whether the obtained point has already been given depending on whether a user ID is registered in association with the point ID received together with the point request. When the point has already been given, the process proceeds to step S**35**. When the point has not been given, the process proceeds to step S**36**.

Step S35

[0134] When the determination unit **306** determines that the point has already been given, the determination unit transmits a point error indicating that the point has already been given, to the consumer terminal **20** via the communication unit **301**. The communication unit **201** of the consumer terminal **20** receives the point error. The display control unit **206** of the consumer terminal **20** displays the point error received by the communication unit **201** on the display **815**. In this case, the consumer has failed to earn the point. Then, the point obtaining process ends.

Step S36

[0135] When the determination unit **306** determines that the point associated with the point ID is not an already-given point, the point addition unit **307** of the eco-point management system **30** refers to the user management DB **310***a*, and adds the point associated with the point ID to the point balance associated with the user ID of the consumer who uses the consumer terminal **20** that has transmitted the point request to the communication unit **301**. Thus, the point addition unit **307** updates the point balance.

Step S37

[0136] The point addition unit **307** notifies the added points and the result of addition of points (updated point balance) to the consumer terminal **20** via the communication unit **301**. The communication unit **201** of the consumer terminal **20** receives the added points and updated point balance. Further, the point addition unit **307** registers the user ID of the consumer who uses the consumer terminal **20** in association with the point ID received by the communication unit **301** in the point management DB **310***c* as illustrated in FIG. **12**. This indicates that the point associated with the point ID has been given, and unauthorized use such as double reading of the QR CODE is prevented. The information indicating that the point has been given is not limited to a user ID, but, for example, flag information indicating that the point has been given may be registered.

[0137] Then, the display control unit **206** of the consumer terminal **20** displays the added points received by the communication unit **201** and the updated point balance on the display **815**. In this case, the consumer has successfully earned the point. FIG. **13** (*d*) illustrates an example of the display of the earned points on the display **815** of the consumer terminal **20**. The total number of points (updated point balance) may be displayed on the display **815**. Then, the point obtaining process ends.

[0138] This process is expected to encourage consumers to purchase the goods of the brand owner to earn more points and encourage the printing company to make efforts to reduce carbon dioxide emission to be assigned with higher points and selected as the partner from the brand owner.

Coupon Exchange Process by Information Processing System

[0139] FIG. **14** is a flowchart of a coupon exchange process performed by the information processing system **1**. FIG. **15** is a diagram illustrating changes in the screen display of the consumer terminal **20** in the coupon exchange process performed by the information processing system **1**. The coupon exchange process performed by the information processing system **1** is described with reference to FIGS. **14** and **15**.

Step S51

[0140] The consumer performs an operation to display a list of coupons available to the consumer on the touch panel **818** of the consumer terminal **20** to exchange the points earned by the consumer with a coupon. For example, the consumer displays a "my page" screen illustrated in (b) of FIG. **15** from the top screen of the consumer terminal **20** illustrated in (a) of FIG. **15**, and presses the "exchange point" button.

Step S52

[0141] The coupon list request unit **204** of the consumer terminal **20** transmits a request for the list of coupons available to the consumer to the eco-point management system **30** via the communication unit **201** in response to the operation on the touch panel **818** by the consumer. Then, the communication unit **301** of the eco-point management system **30** receives the request for the list of available coupons.

Step S53

[0142] When the communication unit **301** receives the request for the list of coupons from the consumer terminal **20**, the coupon list obtaining unit **308** of the eco-point management system **30** refers to the coupon-voucher management DB **310***b* and obtains the list of coupons available to the consumer using the consumer terminal **20**. For example, the coupon list obtaining unit **308** obtains a list of coupons that satisfies the expiration date or the condition among the coupons registered in the coupon-voucher management DB **310***b*.

Step S54

[0143] The coupon list obtaining unit **308** transmits the obtained list of available coupons to the consumer terminal **20** via the communication unit **301**. Then, the communication unit **201** of the consumer terminal **20** receives the list of available coupons.

Step S55

[0144] The display control unit **206** of the consumer terminal **20** displays the list of available coupons received by the communication unit **201** on the display **815**. For example, the display control unit **206** controls the display **815** to display the list of the available coupons as illustrated in (d) of FIG. **15**. Alternatively, the screen illustrated in (c) of FIG. **15** displaying the points earned by reading the QR CODE may be transitioned to the screen illustrated in (d) of FIG. **15** displaying the list of coupons.

Step S**56**

[0145] The consumer performs an operation on the touch panel **818** to select a coupon that the consumer desires to exchange with the earned points from the list of available coupons displayed on the display **815** of the consumer terminal **20**.

Step S**57**

[0146] The coupon request unit **205** of the consumer terminal **20** transmits a request to obtain the coupon selected by the consumer from the list of available coupons to the eco-point management system **30** via the communication unit **201**. Then, the communication unit **301** of the eco-point management system **30** receives the request to obtain the coupon.

Step S**58**

[0147] When the communication unit **301** receives the request for obtaining the coupon selected by the consumer from the consumer terminal **20**, the coupon processing unit **309** of the eco-point management system **30** compares the number of points for obtaining the coupon specified by the request with the point balance of the consumer recorded in the user management DB **310***a*. When the number of points for obtaining the coupon is greater than the point balance of the consumer (the possessed points are insufficient), the process proceeds to step S**59**. When the number of points is equal to or less than the points possessed by the consumer (the possessed points are sufficient), the process proceeds to step S**61**.

Step S**59**

[0148] When the number points for obtaining the coupon is larger than the number of points possessed by the consumer, the coupon processing unit **309** transmits an error indicating point insufficiency to the consumer terminal **20** via the communication unit **301**. Then, the communication unit **201** of the consumer terminal **20** receives the error indicating point insufficiency.

Step S**60**

[0149] The display control unit **206** of the consumer terminal **20** controls the display **815** to display the error indicating point insufficiency received by the consumer terminal **20**. Then, the coupon exchange process ends.

Step S**61**

[0150] When the number of points for obtaining the coupon is equal to or less than the points possessed by the consumer, the coupon processing unit **309** sets the coupon to be a possessed coupon by the consumer.

Step S**62**

[0151] The coupon processing unit **309** deletes the coupon processed by the consumer from the list of the available coupons obtained by the coupon list obtaining unit **308**.

Step S**63**

[0152] The coupon processing unit **309** refers to the user management DB **310***a* and subtracts the number of points equivalent to the coupon from the point balance of the consumer.

Step S**64**

[0153] The coupon processing unit **309** transmits information indicating the success of obtaining the coupon to the consumer terminal **20** via the communication unit **301**. Then, the communication unit **201** of the consumer terminal **20** receives the information indicating the success of obtaining the coupon.

Step S**65**

[0154] The display control unit **206** displays, on the display **815**, the information indicating the success of obtaining the coupon received by the consumer terminal **20**. Then, the coupon exchange process ends.

[0155] As described above, in the eco-point management system **30**, the ID registration unit **303** issues a point ID identifying the points with which a consumer obtains a coupon and registers the point ID in the point management DB **310***c*. The point calculation unit **304** calculates the actual carbon dioxide emission from the activity amount of the printed material on which the QR CODE including the point ID is printed and calculates the number of points based on the actual carbon dioxide emission. The point registration unit **305** registers the number of points calculated by the point calculation unit **304** in the point management DB **310***c* in association with the point ID. Even if the printing company makes efforts to reduce carbon dioxide emission in creating printed materials, the printing company is not motivated to be environmentally conscious unless brand owners aware the reduction result. In view of this, the eco-point management system **30** allows the brand owner to recognize that the printed material has created in an environmentally conscious manner, and the environmental awareness of the printing company increases.

[0156] In the description above, when at least one of the functional units of the information processing system **1** is implemented by a program executed by the CPU, the program may be preinstalled in a ROM or any desired memory of the information processing system **1**. Further, the program executed by the devices included in the information processing system **1** may be stored, as a file installable to or executable by a computer, in a computer-readable recording medium, such as a CD-ROM, a flexible disk (FD), a CD-R, or a DVD. Further, the program executed by the information processing system **1** may be stored on a computer connected to a network such as the Internet, to be downloaded via the network. Further, the computer program executed in the information processing system **1** may be provided or distributed as a web application via a network such as the Internet. The program executed by the information processing system **1** has a module configuration including at least one of the above-described functional units. As the hardware, when the CPU reads the program from the memory and executes the program, the above-described functional units are loaded onto the main memory (work area) to operate as the functional units.

Aspects of the Present Disclosure are as Follows.

[0157] In a first aspect, a point management system that manages a point for a consumer to obtain a coupon includes a first registration unit to issue first identification information identifying the point and register the first identification information in first association information (e.g., a point management DB), a calculation unit to calculate an actual value of a carbon dioxide emission from an amount of activity for a printed material bearing a symbol including the first identification information, calculate the number of points based on the actual value, and a second registration unit to register the number of points calculated by the calculation unit in the first association information in association with the first identification information.

[0158] In a second aspect, in the point management system according to the first aspect, the first registration unit issues the first identification information when a job to create the printed material is generated in a printing management system of a printing company, and the point management system further includes a reception unit to receive the amount of activity for the printed material created in the job. The calculation unit calculates the actual value from the amount of activity received by the reception unit.

[0159] In a third aspect, in the point management system according to the first or second aspect, the calculation unit obtains a predicted value of the carbon dioxide emission using a learned model that predicts the carbon dioxide emission from the amount of activity for the printed material, and calculates the number of points based on the predicted value and the actual value.

[0160] In a fourth aspect, in the point management system according to the second aspect, the reception unit further receives the first identification information included in the symbol read by a first information processing terminal used by the consumer, and the point management system further includes a determination unit to determine whether the first association information indicates that the point associated with the first identification information received by the reception unit has already been given.

[0161] In a fifth aspect, the point management system according to the fourth aspect further includes an addition unit. When the determination unit determines that the point has not yet been given, the addition unit add the point to the point balance of the consumer in management information for managing the point balance of the consumer, and register the point and second identification information (e.g., consumer ID) identifying the consumer in the first association information in association with each other.

[0162] In a sixth aspect, in the point management system according to the fifth aspect, the determination unit determines that the point has been given when the second identification information associated with the point is registered in the first association information.

[0163] In a seventh aspect, the point management system according to the second aspect further includes a third registration unit that, in response to a request from a second information processing terminal used by a provider who provides the printed material, registers third identification information identifying the provider, the number of points for obtaining the coupon, and the name of a product to which the coupon is applicable in association with each other in second association information (e.g., a coupon-voucher management DB).

[0164] In an eighth aspect, in the point management system according to the seventh aspect, the reception unit receives a coupon obtaining request from a first information processing terminal used by the consumer, and the point management system further includes a processing unit. The processing unit obtains the point balance of the consumer from management information for managing the point balance of the consumer, obtains, from the second association information (e.g., a coupon-voucher management DB), to the number of points associated with the coupon requested by the obtaining request received by the reception unit, and compares the point balance with the number of points. When the number of points is equal to or less than the point balance of the consumer, the processing unit sets the coupon to be a possessed coupon by the consumer.

[0165] In a ninth aspect, in the point management system according to any one of the first aspect to the eighth aspect, the symbol is a QR CODE.

[0166] In a tenth aspect, a point management method for managing points for a consumer to obtain a coupon includes registering first identification information identifying the point in first association information, calculating an actual value of a carbon dioxide emission from an amount of activity for creating a printed material bearing a symbol including the first identification information, calculating the number of points based on the actual value, and registering the points calculated by the calculation unit in the first association information in association with the first identification information.

[0167] In an eleventh aspect, a non-transitory recording medium carries computer readable codes for controlling a computer system to perform a method for managing points for a consumer to obtain a coupon.

[0168] The method includes registering first identification information identifying the point in first association information, calculating an actual value of a carbon dioxide emission from an amount of activity for creating a printed material bearing a symbol including the first identification information, calculating the number of points based on the actual value, and registering the points calculated by the calculation unit in the first association information in association with the first identification information.

[0169] The above-described embodiments are illustrative and do not limit the present invention. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present invention. Any one of the above-described operations may be performed in various other ways, for example, in an order different from the one described above.

[0170] The functionality of the elements disclosed herein may be implemented using circuitry or processing circuitry which includes general purpose processors, special purpose processors, integrated circuits, application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), and/or combinations thereof which are configured or programmed, using one or more programs stored in one or more memories, to perform the disclosed functionality.

Processors are considered processing circuitry or circuitry as they include transistors and other circuitry therein. In the disclosure, the circuitry, units, or means are hardware that carry out or are programmed to perform the recited functionality. The hardware may be any hardware disclosed herein which is programmed or configured to carry out the recited functionality.

[0171] There is a memory that stores a computer program which includes computer instructions. These computer instructions provide the logic and routines that enable the hardware (e.g., processing circuitry or circuitry) to perform the method disclosed herein. This computer program can be implemented in known formats as a computer-readable storage medium, a computer program product, a memory device, a record medium such as a CD-ROM or DVD, and/or the memory of an FPGA or ASIC.

1. A point management system comprising circuitry configured to:
- issue first identification information identifying a point for a consumer to obtain a coupon;
- register the first identification information in first association information;
- calculate an actual value of a carbon dioxide emission from an amount of activity for a printed material, the printed material bearing a symbol including the first identification information;
- calculate a number of points based on the actual value; and
- register the calculated number of points in the first association information in association with the first identification information.

2. The point management system according to claim **1**, wherein the circuitry is configured to:
- issue the first identification information in response to a registration request for the first identification information, the registration request being transmitted from a printing management system of a provider of the printed material when the printing management system generates a job to create the printed material; and
- receive the amount of activity for the printed material in the job from the printing management system.

3. The point management system according to claim **1**, wherein the circuitry is configured to:
- obtain a predicted value of the carbon dioxide emission using a learned model that predicts the predicted value from the amount of activity for the printed material; and
- calculate the number of points based on the predicted value and the actual value.

4. The point management system according to claim **2**, wherein the circuitry is configured to:
- receive the first identification information included in the symbol read by a first information processing terminal used by the consumer; and
- determine whether the first association information indicates that one or more points associated with the received first identification information have already been given.

5. The point management system according to claim **4**, wherein the circuitry is further configured to:
- add the one or more points to a point balance of the consumer in management information when determining that the one or more points have not yet been given, the management information recording the point balance of the consumer; and
- register the one or more points not yet given and second identification information identifying the consumer in the first association information in association with each other.

6. The point management system according to claim **5**, wherein the circuitry is configured to:
- determine that the one or more points have already been given when the second identification information associated with the one or more points is registered in the first association information.

7. The point management system according to claim **2**, wherein the circuitry is configured to:
- in response to a request from a second information processing terminal used by a provider of the printed material, register third identification information identifying the provider, a number of points for obtaining the coupon, and a name of a product to which the coupon is applicable in association with each other in second association information.

8. The point management system according to claim **7**, wherein the circuitry is configured to:
- receive a coupon obtaining request from a first information processing terminal used by the consumer;
- obtain a point balance of the consumer from management information recording the point balance of the consumer;
- obtain a number of points associated with the coupon requested by the coupon obtaining request from the second association information;
- compare the point balance with the number of points; and
- set the coupon as being possessed by the consumer when the number of points is equal to or less than the point balance of the consumer.

9. The point management system according to claim **1**, wherein the symbol is a QR CODE.

10. A point management method comprising:
- issuing first identification information identifying a point for a consumer to obtain a coupon;
- registering the first identification information in first association information;
- calculating an actual value of a carbon dioxide emission from an amount of activity for a printed material, the printed material bearing a symbol including the first identification information;
- calculating a number of points based on the actual value; and
- registering the calculated number of points in the first association information in association with the first identification information.

11. A non-transitory recording medium storing a plurality of program codes which, when executed by one or more processors, causes the one or more processors to perform a method, the method comprising:
- issuing first identification information identifying a point for a consumer to obtain a coupon;
- registering the first identification information in first association information;
- calculating an actual value of a carbon dioxide emission from an amount of activity for a printed material, the printed material bearing a symbol including the first identification information;

calculating a number of points based on the actual value; and

registering the calculated number of points in the first association information in association with the first identification information.

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